

The *Volta* Confluence Engine

THE WORLD'S FIRST NON-ELECTRIC TWO-CHANNEL
PHONOGRAPH MIXER

CONCEPT ORIGIN

Jake Galm,
researched &
documented
2026

NAMED FOR

Alessandro Volta
1745–1827

GRID
CONNECTION

None

CHEMICAL
BATTERIES

None

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Philosophy

II Complete System
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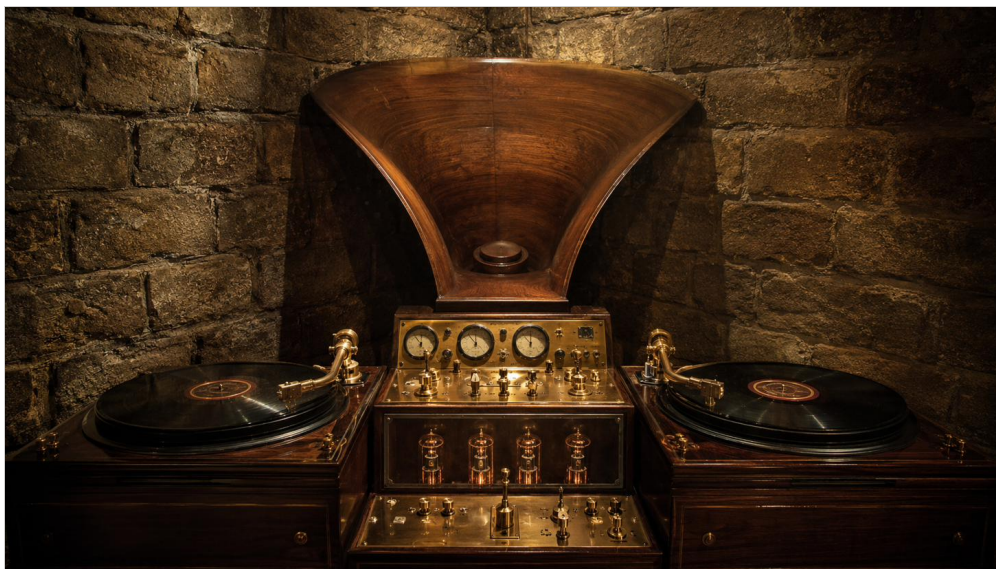
A Standard Hand Crank

B Gear-Assisted Hand
Crank

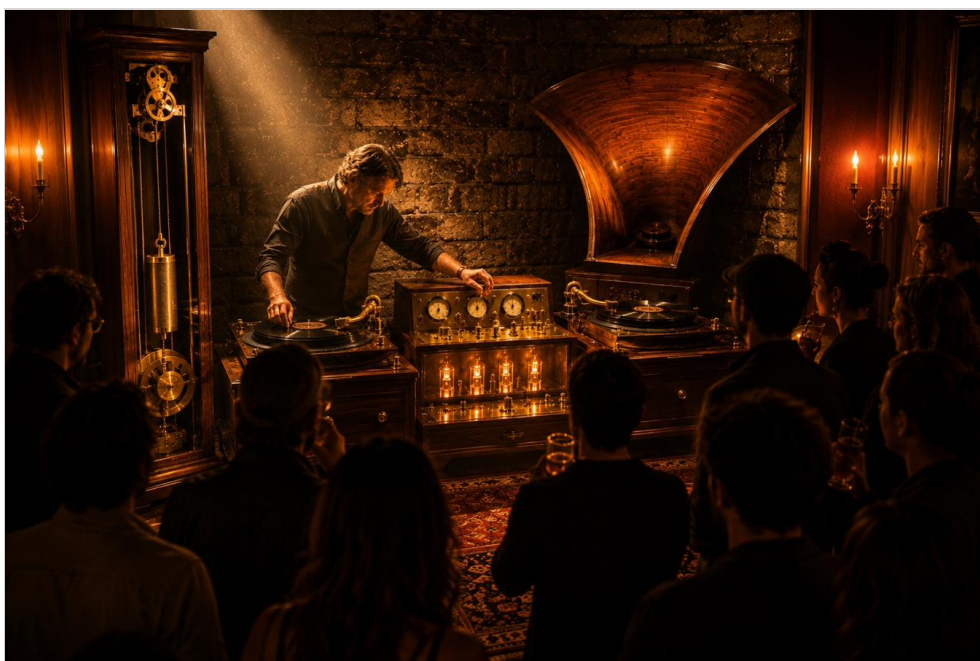
	Bearing System		B+	Peripheral Light Indicator — Add-On for Options A & B
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SECTION I

Instrument Overview & *Philosophy*



IMG-01 — *The Complete Instrument in Performance*



IMG-15 — *Live Performance — Operator and Audience*



IMG-16 — *Live Performance — Wide Audience View*

The Volta Confluence Engine is the world's first two-channel phonograph mixer capable of playing modern vinyl records — 33 $\frac{1}{3}$ and 45 RPM — with independent channel level control, a mechanical crossfader, and master volume, all without connection to an electrical grid and without chemical batteries of any kind.

The instrument draws its name from Alessandro Volta (1745–1827), who understood both electrostatic storage and the electromagnetic principles that followed. The design follows a Victorian technological lineage — every operating principle was either discovered, demonstrated, or theoretically available before 1910. Where modern materials are used, they execute Victorian principles with precision unavailable to Victorian engineers, not concepts unknown to them.

All energy powering the Volta Confluence Engine originates from the operator's own muscle effort, stored mechanically or electrostatically, and released silently on demand. The complete power chain from human effort to acoustic output contains no combustion, no chemical reaction, no grid connection, and no battery chemistry.

★ WHAT THIS INSTRUMENT SOUNDS LIKE

Modern vinyl at full fidelity — 20Hz to 20kHz, flat RIAA equalisation, correct stereo channel separation. Volume capable of filling a large room at 100dB+ SPL at one metre from the horn

mouth. The noise floor is vinyl surface noise — the electronics are quieter than the record itself. Single-ended triode amplification produces 0.3–1% second-harmonic distortion — below the distortion of the vinyl groove. The sound is indistinguishable from a reference-quality valve phono system connected to a high-efficiency horn speaker, because that is precisely what it is, powered entirely by the operator's own physical effort.

DESIGN CONSTRAINTS – FIRM

No connection to the electrical grid. No chemical battery. No combustion engine. Electromagnetism is permitted. All modern materials and manufacturing techniques are available. The instrument must play standard 33⅓ and 45 RPM vinyl records. Two independent channels with crossfader. Concert-capable room volume.

VICTORIAN LINEAGE – COMPONENT BY COMPONENT

COMPONENT / PRINCIPLE	FIRST AVAILABLE	HISTORICAL BASIS	MODERN ENHANCEMENT
Leyden jar / capacitor storage	1745	Van Musschenbroek, von Kleist	Barium titanate ceramic dielectric — 1,000× better than glass
Electromagnetic generation	1831	Faraday's dynamo principle	Neodymium magnets, precision windings
Voltage multiplication	1840s concept	Capacitor + rectifier theory	Cockcroft-Walton ladder — 1932 execution
Moving coil transducer	1877	Werner von Siemens	Diamond stylus, modern cartridge geometry
Remontoire spring mechanism	1739	John Harrison, H2 chronometer	Beryllium copper spring, sapphire

			bearings
Eddy current braking	1855	Léon Foucault	Precision copper disc, N52 neodymium
Thermionic valve / triode	1906	De Forest Audion; emission 1883	Modern 12AX7, stable high-voltage supply
Field coil loudspeaker	1925	Rice and Kellogg, Bell Labs	Custom wound for high-voltage DC supply
Exponential horn speaker	1919	Webster acoustic horn equation	CNC-accurate tractrix profile
Weight-driven mechanism	13th century	Tower clocks, grandfather clocks	Precision guide rails, Dyneema safety

SECTION II

Complete System *Architecture*

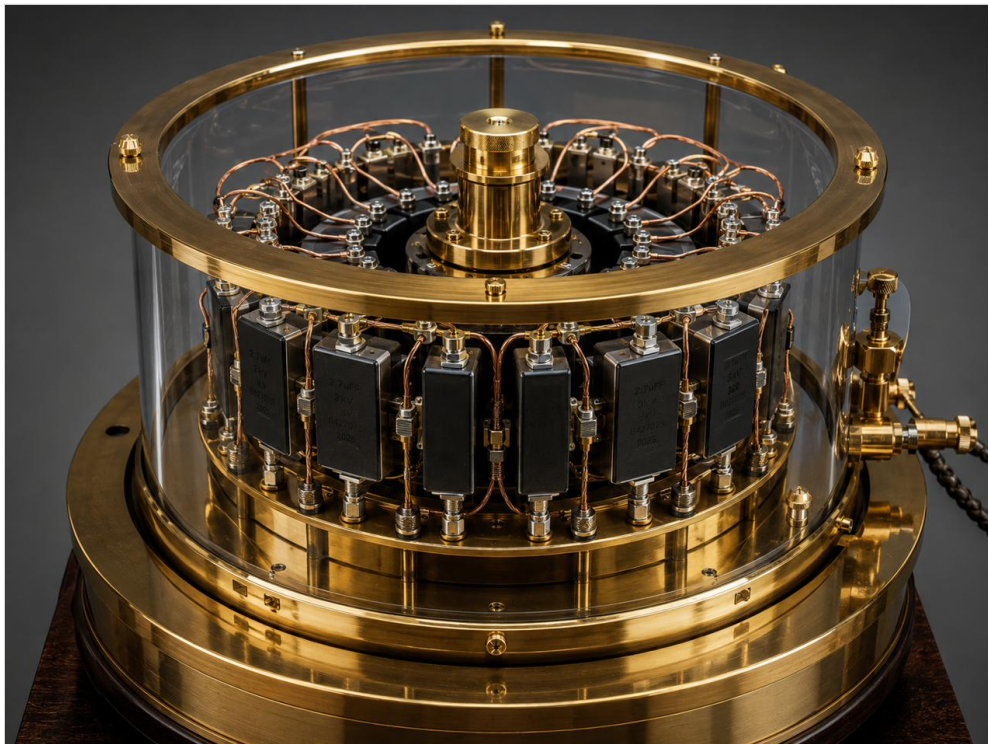
The system has five functional layers. Each layer is electrically and mechanically independent from the others wherever possible. Failures in one layer do not cascade.

LAYER 1 – HUMAN ENERGY INPUT

The operator provides mechanical energy through one of three methods detailed in Section VIII. This energy is converted to rotary motion at a flywheel-equipped generator shaft in the base cabinet.

LAYER 2 – HIGH-VOLTAGE ELECTROSTATIC STORAGE

The generator produces 24V AC via electromagnetic induction. A Cockcroft-Walton voltage multiplier ladder — entirely passive diodes and capacitors — steps this to 600V DC. This charges a bank of high-voltage ceramic capacitors. The bank stores 4,000–5,000 joules. Self-discharge is negligible on performance timescales. This is the instrument's primary electrical energy reservoir — its modern Leyden jar battery.



IMG-09 — The Capacitor Bank — Modern Leyden Jars

LAYER 3 – POWER DISTRIBUTION

The 600V storage rail supplies all electronic circuits through passive resistive voltage dividers and zener reference stages. No switching regulators. No DC-DC converters. No switching noise. Every supply rail is derived passively from the high-voltage source.

LAYER 4 – SIGNAL CHAIN

Two independent phono preamplifier chains, each consisting of passive step-up transformer, two triode valve gain stages with passive RIAA equalisation network between them, transformer volume control, and mutual inductance crossfader. Powered from Layer 3. Output to field coil horn drivers.

LAYER 5 – TURNTABLE MECHANICS

Two turntables, each independently driven by a precision remontoire

spring motor. Completely independent of all electrical systems — the platters spin from stored spring energy alone. The spring motor has no electromagnetic components. The eddy current speed governor requires no power source — it is a passive electromagnetic braking system.

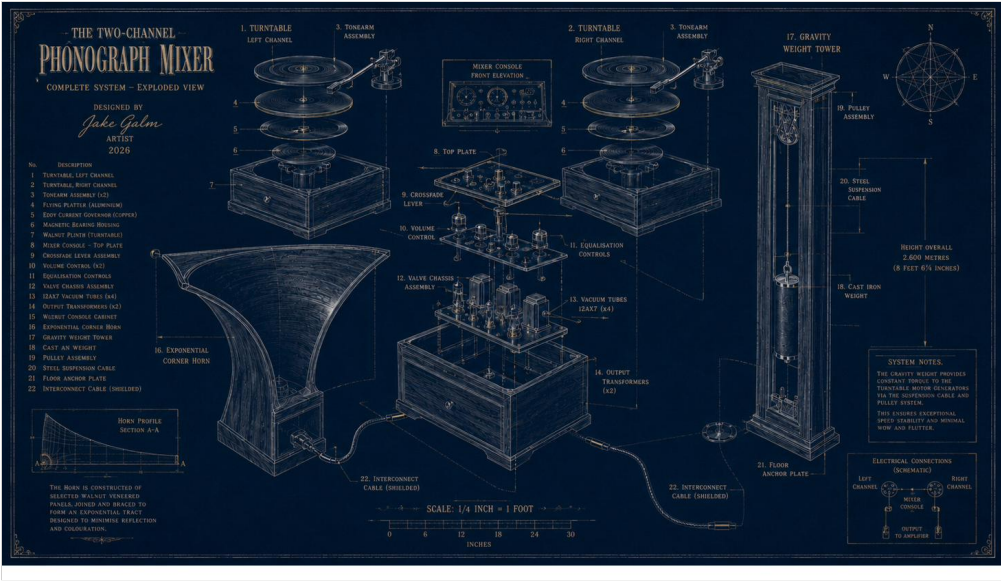
SYSTEM BLOCK DIAGRAM - ENERGY FLOW

HUMAN EFFORT (muscle energy) | ▼ GENERATOR FLYWHEEL (kinetic buffer) | 24V AC ▼ COCKCROFT-WALTON MULTIPLIER (passive) | 600V DC ▼ CERAMIC CAPACITOR BANK (4,000-5,000 J) |

▼ ▼ ▼ ▼ VALVE HEATERS PHONO PREAMPS FIELD COIL PLL SPEED 6.3V / 7.5W 100V / 2W per ch SPEAKERS 250V REFERENCE (via divider) (via dropping R) 8-13W total 12V / 30mW | |

| ▼ SIGNAL PATH MC CARTRIDGE → SUT → V1 → RIAA → V2 → TVC → CROSSFADER | FIELD COIL DRIVERS | EXPONENTIAL HORN | ROOM

TURNTABLE MECHANISM (independent of all above): MAINSPRING → REMONTOIRE → PLATTER BEARING (magnetic) | EDDY CURRENT GOVERNOR (no power required)



IMG-08 (revised) — Exploded View — Victorian Engineering Blueprint

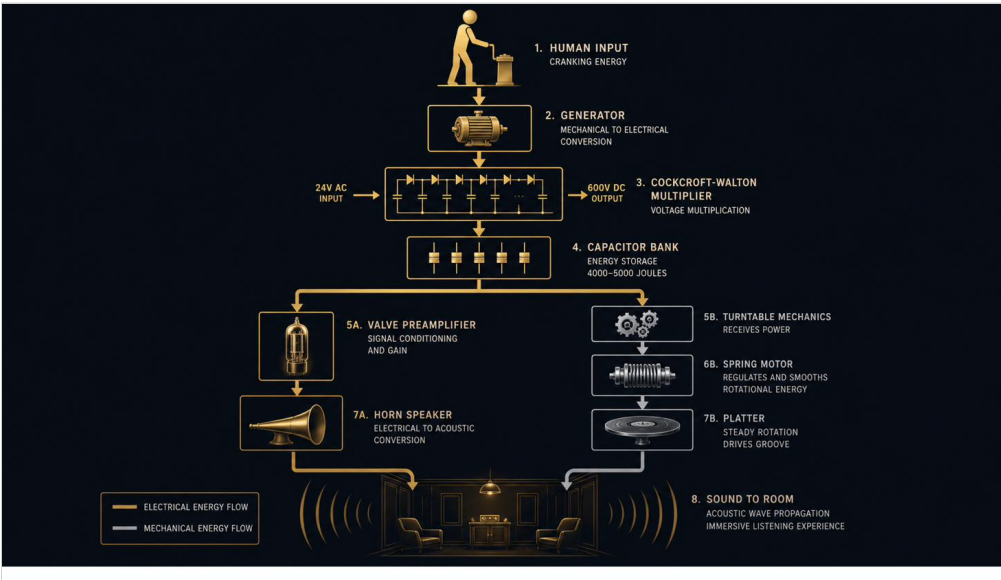
POWER BUDGET SUMMARY

CONSUMER	VOLTAGE	CURRENT	POWER	SOURCE
Valve heaters (×4 triodes)	6.3V DC	1.2A total	7.5W	Cap bank via divider

Phono preamps (×2 channels)	100V DC	10mA each	2W total	Cap bank via dropping R
Field coil comp. drivers (×2)	250V DC	20–30mA each	5–8W	Cap bank via dropping R
Field coil woofer (×1)	250V DC	10–20mA	3–5W	Cap bank via dropping R
PLL speed reference (×2)	12V DC	2.5mA each	0.06W	Cap bank via zener
Total continuous	—	—	18–23W	Capacitor bank

ENERGY BUDGET NOTE

At 20W continuous draw, a 5,000-joule fully charged capacitor bank provides 250 seconds — approximately 4 minutes — of operation before requiring replenishment from the generator. The operator must provide energy to the generator every 3–4 minutes throughout the performance. The method of providing this energy — and its implications — is the subject of Section VIII.



IMG-14 — Energy Flow — System Diagram

SECTION III

Turntable *Specification*

Both turntables are identical in specification. They are mechanically independent — no shared components, no mechanical coupling between them. Each is a self-contained playback instrument.

Platter Bearing — Neodymium Magnetic Bearing

The platter bearing is the single most critical mechanical component in the turntable. Conventional bearings contribute rumble — low-frequency mechanical noise that the cartridge reads as signal. The Volta Confluence Engine uses a neodymium magnetic bearing, eliminating bearing contact entirely.



IMG-03 — *The Floating Platter — Turntable Detail*

MAGNETIC BEARING — CROSS SECTION SCHEMATIC

PLATTER | ← Machined aluminium

alloy | (310mm dia) | 2.5kg minimum mass

_____ ||

LOWER PLATTER RING || ← N52 neodymium ring magnets || (repelling, N-N facing) || Outer dia: 80mm _____

Inner dia: 60mm | Field-shielded configuration

_____ || UPPER PLINTH RING || ← Fixed to plinth, same polarity || (repelling, N-N facing) || Mu-metal containment ring _____ prevents upward field escape | CERAMIC SHAFT ← 6mm diameter, polished (lateral constraint only – zero vertical contact) |

_____ | PLINTH |

_____ Platter floats 1.2–1.8mm above bearing surface. Bearing friction: effectively zero. Rumble contribution: below measurement threshold. Field shielding mandatory – N52 magnets adjacent to MC cartridge without shielding will induce hum in signal chain.

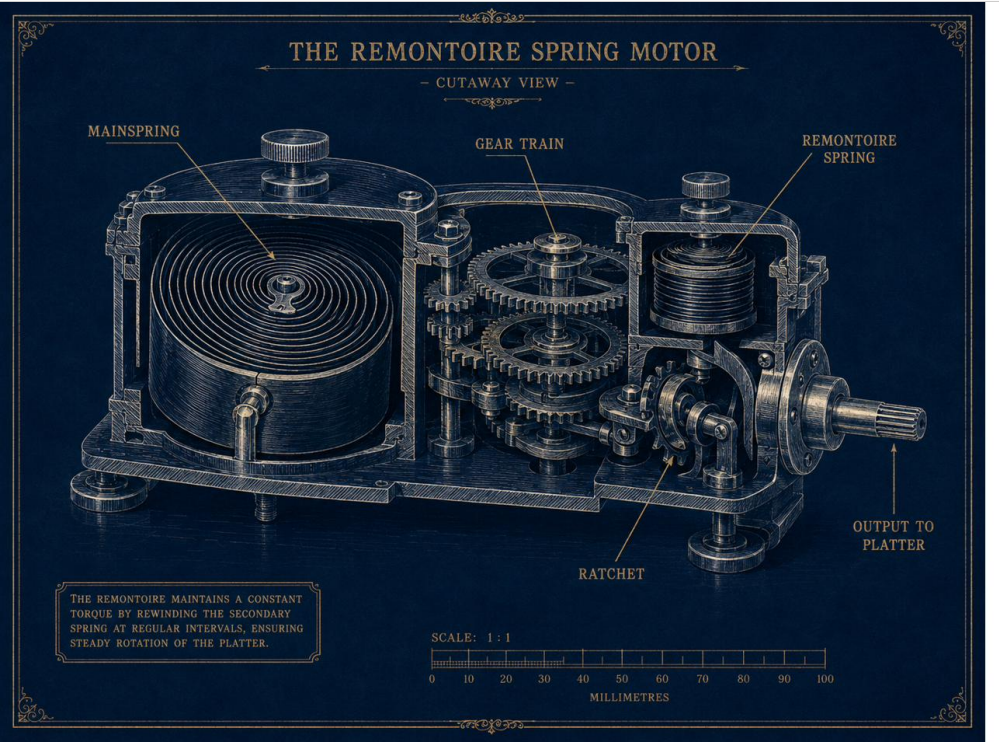
BEARING SPECIFICATION

PARAMETER	VALUE	NOTES
Bearing type	Neodymium magnetic repulsion bearing	Clearaudio CMB principle — may purchase or fabricate
Magnet grade	N52 neodymium iron boron	Highest available energy product
Ring dimensions	OD 80mm, ID 60mm, height 12mm each	Two rings, same polarity facing
Float height	1.2–1.8mm nominal	Set by ring spacing during installation
Lateral constraint	6mm polished alumina ceramic shaft	No vertical contact, minimal lateral friction
Field shielding	Mu-metal containment ring, 0.5mm thick	Mandatory — prevents cartridge interference

Rumble specification	<-70dB DIN 45539	Expected based on Clearaudio CMB measured performance
Platter mass	2.5kg minimum	Higher mass improves speed stability
Platter material	6061 aluminium alloy	CNC machined, hand-balanced to <1g.cm imbalance

Drive System — Remontoire Spring Motor

Each platter is driven by a precision spring motor with a remontoire (constant-force) mechanism. The remontoire — invented by John Harrison in 1739 for his H2 marine chronometer — decouples the platter from variations in the mainspring's tension as it unwinds, delivering constant torque regardless of how much energy remains in the spring.



IMG-11 — Spring Motor — Remontoire Mechanism

REMONTOIRE SPRING DRIVE - FUNCTIONAL SCHEMATIC

MAINSRING → GEAR TRAIN → REMONTOIRE → PLATTER SPINDLE
(energy (reduction) (constant (driven at storage) force stage)
33⅓ RPM) REMONTOIRE DETAIL: Main spring drives a SMALL SECONDARY

SPRING via ratchet. Secondary spring is always wound to IDENTICAL TENSION. Platter receives drive ONLY from secondary spring. Main spring rewinds secondary spring every 8-10 seconds. Result: Platter sees constant torque regardless of main spring state. Torque variation to platter: <2%. MAINTAINING POWER: During manual winding of mainspring, a THIRD small spring (maintaining power spring) continues driving platter. Platter never stops or hesitates during winding. Record plays through without interruption. SPEED SELECTION: 33 $\frac{1}{3}$ RPM — Standard LP position (default) 45 RPM — Stepped pulley, position 2 78 RPM — Stepped pulley, position 3 (for collectors)

SPRING MOTOR SPECIFICATION

PARAMETER	VALUE	NOTES
Mainspring material	High-carbon spring steel, precision ground	Commission from specialist spring manufacturer
Mainspring dimensions	Strip width 25mm, thickness 0.6mm, active length 1,200mm	Custom wound to specification
Barrel diameter	180mm outer	Larger than watch spring — tower clock scale
Remontoire spring	Beryllium copper, 40mm × 0.3mm	Consistent elastic modulus, fatigue resistant
Remontoire cycle	8–10 seconds	Each cycle: secondary spring rewinds from mainspring
Winding	Hand crank, 40 turns per side	Approximately 3 minutes. Maintaining power allows mid-play winding.
Run time per wind	25–28 minutes	Sufficient for one LP side. Wind between sides.

Ratchet	48-tooth hardened steel, sapphire pawl	One-way engagement, prevents backwind
Commission source	Precision clockmaker or specialist spring motor manufacturer	Not standard catalogue item — must be custom built
Estimated cost per unit	\$3,000–\$7,000	Major cost driver. Clockmaker commission varies widely.

Speed Regulation — Eddy Current Governor

Speed is regulated by an eddy current governor — a passive electromagnetic device requiring no power source. As the platter spins, a polished copper disc attached to the platter spindle rotates near a neodymium horseshoe magnet mounted on the plinth. The rotating disc induces eddy currents that oppose its motion, applying a braking force proportional to speed. Faster rotation produces stronger braking; slower rotation produces weaker braking. This self-regulating behaviour maintains constant speed without any active control circuit.

A crystal-referenced Phase-Locked Loop (PLL) circuit — powered from the capacitor bank at 12V, consuming 30mW — compares actual platter speed against a quartz crystal reference and fine-adjusts the eddy current braking load resistance to maintain speed within 0.05% of target. Wow and flutter: below 0.05% WRMS.

SPEED GOVERNOR SPECIFICATION

PARAMETER	VALUE	NOTES
Governor type	Eddy current brake + PLL correction	Passive primary + active fine-trim
Copper disc	200mm diameter, 6mm thick, OFHC copper	OFHC (oxygen-free high-conductivity) for maximum conductivity
Braking magnet	N52 neodymium	Set during installation;

	horseshoe, adjustable proximity	locked with grub screw
PLL reference	Quartz crystal oscillator, 32.768kHz	Post-1921 technology. Alternative: tuning fork at 0.15% W&F
PLL power consumption	30mW at 12V	Negligible fraction of total system draw
Achieved W&F (with PLL)	<0.05% WRMS	Better than most commercial belt-drive turntables
Achieved W&F (tuning fork)	<0.15% WRMS	If strict Victorian authenticity preferred
Speed calibration	Set once during installation	Requires strobe disc and 50Hz or 60Hz reference

Tonearm & Cartridge

PARAMETER	SPECIFICATION	NOTES
Arm tube	Carbon fibre, 9.5-inch effective length	Low mass, high stiffness, non- resonant
Pivot bearing	Unipivot — sapphire needle on polished tungsten carbide cup	Zero friction, slight lateral sway — known limitation
Anti-skating	Thread and weight, adjustable	Conventional. Magnetic anti-skate also viable.
Headshell	Magnesium alloy, 8g	Low mass for correct tonearm resonance frequency
Cartridge type	Low-output moving coil (LOMC)	0.2–0.5mV output. Best available within budget.

Stylus profile	Line contact or microline diamond	Superior groove tracing, reduced distortion
Tracking force	Per manufacturer specification, typically 1.8–2.5g	Set with precision stylus force gauge
Recommended cartridges	Denon DL-103R, Ortofon MC Cadenza, Lyra Delos	All excellent LOMC options at different price points
Internal wiring	Silver-alloy Litz wire, 4-conductor shielded	Twisted pair for each channel, separate shield

SECTION IV

Signal Chain — *Complete*

The complete signal chain from groove to horn is described below. This chain is identical for both channels until the crossfader stage, where they are combined.

SIGNAL CHAIN – PER CHANNEL

STAGE 1: MC CARTRIDGE Output: 0.2–0.5mV, 3–10 ohm source impedance Energy source: groove modulation only (passive) ↓
STAGE 2: PASSIVE STEP-UP TRANSFORMER (SUT) Turns ratio: 1:10 Voltage gain: +20dB Mounted: Mu-metal shielded brass housing, on tonearm pillar Component: Lundahl LL1931, 1:10 (specified) Cost: \$300–\$700 per unit ↓
STAGE 3: FIRST TRIODE GAIN STAGE Device: 12AX7 (ECC83) dual triode, one section per channel Operating point: 250V anode supply, 1.2mA quiescent Gain: +26dB (×20 voltage) Power: ~3V from 600V rail via dropping resistor Coupling: transformer coupled to RIAA network ↓
STAGE 4: PASSIVE RIAA EQUALISATION NETWORK Type: RC network, IEC 60098 compliant

Components: Metal film resistors 0.1% tolerance, polypropylene caps
Time constants: 3180 μ s / 318 μ s / 75 μ s (standard RIAA)
Attenuation: approximately -20dB at 1kHz (recovered by Stage 5)
Accuracy: $\pm$ 0.5dB 20Hz-20kHz (achievable with 0.1% components) ↓
STAGE 5: SECOND TRIODE GAIN STAGE Device: 12AX7, second section
Operating point: identical to Stage 3 Gain: +26dB Output level:
300-500mV line level, flat 20Hz-20kHz Total gain from cartridge:
approximately +52dB ↓ STAGE 6: TRANSFORMER VOLUME CONTROL (TVC)
Type: Multi-tap toroidal transformer, 24 switched positions
Attenuation range: 0dB to -46dB in 2dB steps Switching:
precision 24-position rotary switch, silver contacts Noise
contribution: effectively zero (passive transformer) ↓ (×2
channels merge at crossfader) STAGE 7: TRANSFORMER TAP
CROSSFADER Type: Complementary dual TVC, mechanically ganged
Law: Equal-power (-3dB both channels at centre) Positions: 24
discrete steps Mechanism: Single lever through precision brass
bellcrank linkage ↓ STAGE 8: MASTER TRANSFORMER VOLUME CONTROL
Type: Single TVC on summed output bus Positions: 12 discrete
steps Control: Large weighted brass knurled collar ↓ STAGE 9:
OUTPUT TRANSFORMER Type: Interstage / output, 1:1 ratio with
impedance matching Purpose: Impedance matching valve output to
field coil driver ↓ STAGE 10: FIELD COIL COMPRESSION DRIVERS +
HORN

RIAA EQUALISATION – WHY IT MATTERS

Modern vinyl records are cut with the RIAA equalisation curve — bass frequencies are reduced during cutting to prevent groove overcrowding, and treble is boosted to improve signal-to-noise. During playback this must be precisely reversed. Without correct RIAA equalisation, vinyl sounds thin, shrill, and bass-free. Every phono preamplifier in the world applies this curve. The passive RIAA network between the two valve gain stages achieves this accurately and passively, with no active components and no additional power draw.

The placement between two gain stages — rather than at the input — is the correct engineering practice: there is enough signal level at this point that the network's inherent attenuation does not compromise signal-to-noise ratio. This is the architecture used in reference valve phono stages since the 1950s.

SECTION V

Mixer *Console*

CONSOLE CABINET SPECIFICATION

PARAMETER	SPECIFICATION
Outer dimensions	700mm W × 500mm D × 350mm H
Cabinet material	18mm Baltic birch plywood, walnut veneer
Control panel	4mm polished brass sheet, machine-engraved labels, lacquered
Internal structure	Mu-metal partition between generator section and audio section
Valve display	Recessed glass-fronted valve bay, warm amber illumination from valve glow
Gauge panel	Three glycerin-filled brass gauges: Cap Bank Charge, Channel A, Channel B
Cable routing	All signal cables separated from power runs by minimum 50mm



IMG-02 — Console Control Panel

CONTROLS – COMPLETE LIST

CONTROL	TYPE	MECHANISM	FEEL
Channel A Volume	TVC fader	24-position rotary, silver contacts	Firm detented clicks, 120° total rotation
Channel B Volume	TVC fader	As above	As above
Crossfader	Dual complementary TVC	Linear lever, 200mm throw, brass bellcrank	Smooth, slight resistance, natural centre feel
Master Volume	TVC on output bus	12-position, large weighted collar	Substantial, satisfying — 80mm diameter collar
Speed Select A	3-position toggle	33 / 45 / 78 RPM — stepped	Positive brass toggle lever

		pulley selector	
Speed Select B	3-position toggle	As above	As above
Winding Crank A	Folding hand crank	Retractable, 18:1 internal reduction (Option B)	Smooth rotation, light resistance
Winding Crank B	Folding hand crank	As above	As above
EQ A (optional)	3-position toggle per channel	Passive LC shelving network	Bass / Flat / Treble
Generator Crank	External crank or gravity weight (Section VIII)	Depends on energy option chosen	—

GAUGES & INDICATORS

GAUGE	TYPE	RANGE	PURPOSE
Capacitor Bank Charge	Glycerin-filled, 63mm, brass bezel	0–100% (0–600V DC)	Primary power reserve indicator. Watch this gauge.
Spring Tension A	Mechanical, driven by mainspring cam	Red / Amber / Green zones	Wind when pointer reaches Amber
Spring Tension B	As above	As above	As above
Valve warm-up	LED indicator — one exception to no-electricity rule	Off / Warm-up / Ready	Powered from capacitor bank. Extinguishes at operating temperature.



SECTION VI

Valve Amplification *System*

The amplification system uses thermionic triode valves — a technology whose operating principle (thermionic emission) was described by Edison in 1883 and whose practical application as an amplifying device dates to De Forest's Audion of 1906. This is the most Victorian-authentic electronic amplification technology available, and it is also, by the consensus of the audiophile community, among the finest sounding.

Single-ended Class A triode amplification produces predominantly second-harmonic distortion — musically consonant, subjectively pleasant, and at levels of 0.3–1% which are below the distortion inherent in the vinyl groove itself. The frequency response is flat within $\pm 0.5\text{dB}$ from 20Hz to 20kHz. The output impedance is matched to the field coil drivers through the output transformer.



IMG-04 — *The Valve Bay*

VALVE SPECIFICATION

PARAMETER	VALUE	NOTES
Valve type	12AX7 (ECC83) dual triode	Available since 1947. Widely available NOS and new production.
Quantity	4 valves total (2 per channel, 1 section per stage)	Buy matched pairs. Keep 4 spares minimum.

Heater voltage	6.3V DC (not AC — eliminates heater hum)	DC heater supply from capacitor bank via regulated divider
Heater current	300mA per valve, 1.2A total	7.5W heater load
Anode voltage	100–250V depending on stage	From 600V rail via precision dropping resistors
Quiescent current	1.0–1.5mA per stage	Set by cathode resistor during commissioning
Warm-up time	60–90 seconds to operating temperature	Do not apply signal until warm-up indicator clears
Recommended brands	Telefunken ECC83, Mullard CV4004, Brimar 12AX7	NOS (new old stock) preferred. GE or RCA acceptable.
Valve retainers	Machined brass ring, friction fit	Prevents vibration-induced microphony

⚠ HIGH VOLTAGE SAFETY

The capacitor bank operates at 600V DC. The valve anode circuits contain voltages up to 250V DC. These voltages are lethal at the current levels available from the capacitor bank. All high-voltage circuits must be fully enclosed within the console cabinet with interlocked panels — opening any panel must disconnect the high-voltage supply. The capacitor bank must include a bleed resistor (1M Ω , 5W) to discharge stored energy within 60 seconds of disconnection. Safety disclaimer must be prominently displayed on the instrument. Builder must be qualified to work with high-voltage electronics or commission a qualified electronics technician for this stage.

THERMOELECTRIC HEATER SUPPLEMENT

Each valve envelope reaches approximately 180–220°C during operation. Bismuth telluride thermoelectric modules bonded to the valve retainer rings harvest this waste heat and contribute 50–150mW per valve to the

heater supply — reducing the net draw on the capacitor bank from 7.5W to approximately 5–6W for the heater circuit. This is not essential but reduces the cranking frequency required during performance. The Seebeck effect — discovered by Thomas Johann Seebeck in 1821 — is the operating principle. Entirely Victorian in lineage.

SECTION VII

Output System — *Horn & Speakers*

Field Coil Compression Drivers

The output drivers use electromagnets — field coils — rather than permanent magnets to generate the magnetic field in which the voice coil moves. Field coil speakers were the standard from the 1920s through the 1940s before permanent magnets became cheap. Audiophiles have rediscovered them because a field coil powered by stable DC produces a stronger, more consistent magnetic field than any permanent magnet of comparable size, resulting in higher sensitivity and a unique dynamic character described by experienced listeners as having "more energy" particularly on transients and crescendi.

The field coil current — approximately 20–30mA at 250V — comes from the capacitor bank through a dropping resistor. The field coil's own inductance (typically thousands of henries at high turn counts) naturally filters ripple from the supply rail, protecting the voice coil from any power supply noise without additional filtering.

SPEAKER SPECIFICATION

PARAMETER	VALUE	NOTES
Compression drivers	Custom field coil, 44mm beryllium diaphragm × 2	Must be custom wound — not available off shelf. Commission from specialist.

Field coil supply	250V DC, 20–30mA	From capacitor bank. Field coil inductance filters noise.
Field strength in gap	1.8–2.2 Tesla	Exceeds any practical permanent magnet
Driver sensitivity	108–112 dB/W/m	Among highest achievable from any driver type
Frequency range (driver)	300Hz–18kHz	Matched with woofer below crossover point
Woofer	15-inch field coil, custom wound	Bass to 30Hz in corner-loaded configuration
Voice coil impedance	8 or 16 ohm (specify at order)	Match to output transformer secondary
Hum bucking coil	Included in driver winding	Historical technique — reduces field coil supply hum
Commissioning source	Specialist field coil speaker builder	Feastrex (Japan), Audio Note (UK), or individual commission
Cost estimate (per driver)	\$1,500–\$4,000	Wide range. Beryllium diaphragm adds significant cost.
Alternative	Premium permanent magnet compression drivers	Reduces cost by \$3,000–\$8,000. Slight efficiency reduction.

Corner-Loaded Exponential Horn



IMG-06 — *The Corner Horn in Situ*

⚠ VENUE REQUIREMENT – CRITICAL

The corner horn requires placement in a true 90° corner where two walls meet the floor. The room corner acts as an acoustic extension of the horn, providing bass loading down to approximately 30Hz. Without a suitable corner, bass response is significantly compromised. This instrument requires venue-appropriate permanent installation. It cannot be moved between venues and sound the same. For touring applications, specify a direct-radiating exponential horn without corner loading, accepting reduced bass

extension below approximately 100Hz.

PARAMETER	VALUE	NOTES
Horn type	Corner-loaded exponential, dual-throat at same vertical axis	Both channels enter at same throat position — single acoustic source
Enclosure dimensions (approx)	900mm W × 700mm D × 1,200mm H	Exact dimensions from acoustic calculation for venue
Wall construction	18mm Baltic birch, CNC-routed internal baffles	Internal damping with bituminous mat
Throat diameter	25mm per channel, Y-confluence inside enclosure	Channels mix acoustically inside horn — no electrical summing
Mouth dimensions	800mm × 600mm minimum	Larger mouth → lower bass extension
Expected sensitivity	100–106 dB/W/m system	Includes horn loading factor
Expected frequency response	30Hz–18kHz	±3dB in suitable corner
Finish	Walnut veneer, satin lacquer, brass hardware	Matches turntable plinths and console aesthetic
Weight	80–110kg assembled	Permanent installation recommended
Cabinetmaker commission	\$3,000–\$6,000	Requires acoustic woodworking experience

Vibration Isolation Platform

Each turntable sits on a five-layer isolation platform. Bottom to top: four

polished brass cups containing precision steel ball bearings (lateral isolation) → EAR Isodamp C-1002 constrained layer bonded to brass cup plate → four precision compression springs rated for 1.5Hz resonant frequency under load (low-frequency isolation) → 50mm Welsh slate slab → turntable plinth on three sorbothane hemispheres of correct durometer. This five-stage system decisively eliminates acoustic feedback between horn and stylus and substantially attenuates floor-borne vibration from any source.

SECTION VIII

Energy Systems — *Three Options*

The instrument requires the operator to supply mechanical energy to the generator throughout the performance. Three methods are offered. Each has different implications for confidence, ergonomics, visual presentation, and venue requirements. The electronic system — valves, phono stages, field coil speakers — is identical across all three. Only the generator drive changes.

ON CONFIDENCE FIGURES

Confidence percentages used throughout this document are engineering judgement estimates based on design analysis, stress testing on paper, and documented failure mode assessment. No physical prototype of the Volta Confluence Engine has been built. All figures should be understood as informed estimates, not measured results. A first-prototype confidence of 88% means the design has been rigorously stress-tested and no fundamental physics objections remain — it does not mean the instrument has been built and tested eight times in ten successfully. The numbers are meaningful as comparative indicators and as a record against which a future builder can assess the quality of this analysis. They are not guarantees.



Standard Hand Crank

DIRECT DRIVE – ZERO ADDED
COMPONENTS – BASELINE DESIGN

93%

2ND PROTOTYPE CONFIDENCE

A single hand crank attached directly to the flywheel generator shaft. The operator rotates the crank at a comfortable pace — approximately 2 revolutions per second — for 60–90 seconds every 3 minutes throughout the performance. Between crank cycles, both hands are free for mixer operation.

MECHANISM

The crank is a folding brass handle on a steel crank arm, mounted on the generator shaft extension protruding from the rear of the console cabinet. A one-way ratchet prevents the crank from being driven by the generator when not cranking. The generator flywheel — a cast iron disc, 200mm diameter, 3kg — stores kinetic energy between crank strokes, delivering smooth rotational input to the generator armature.

ERGONOMICS

Research confirms that the sustained comfortable one-hand cranking power for 95% of the population is approximately 31 watts — exceeding our 20W system requirement. A 60–90 second crank cycle at comfortable pace produces sufficient energy for the subsequent 3 minutes of operation. The total manual effort over a 2-hour performance is approximately 20–24 minutes of comfortable wrist rotation, distributed in 90-second intervals. No meaningful fatigue results from this pattern.

PROBLEMS

One hand is occupied during each crank cycle. During a complex crossfade, the operator must choose between completing the crank and executing the mix with both hands. In practice this is managed by timing crank cycles to track intros, quiet passages, or moments between active mixing — but it cannot be guaranteed to never conflict.

SPECIFICATION

PARAMETER	VALUE
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Crank handle	Turned brass, 22mm diameter, 120mm long, knurled grip
Crank arm	Mild steel, 180mm radius, 8mm section
Generator connection	Direct shaft coupling, no intermediate gears
One-way ratchet	36-tooth hardened steel, spring-loaded pawl
Mounting position	Rear panel of console, 900mm above floor level
Crank cycle	60–90 seconds every 3 minutes
Additional components	None beyond generator assembly
Additional cost	\$80–\$150 (crank mechanism fabrication)

B Gear-Assisted Hand Crank 93%

3:1 PLANETARY GEAR REDUCTION –
LIGHTER EFFORT, SHORTER CYCLES

2ND PROTOTYPE CONFIDENCE

Identical to Option A but with a precision helical planetary gear reduction unit — 3:1 ratio — between the crank handle and the generator shaft. The operator turns the handle at the same comfortable pace; the generator shaft spins three times faster. Crank effort reduces to approximately one third. The 90-second crank cycle becomes 30 seconds at the same pace, or 90 seconds at substantially reduced effort.

WHY HELICAL PLANETARY – NOT SPUR GEAR

Spur gears mesh with a single-line contact, producing an audible impulse at each tooth engagement. Helical gears mesh progressively along a diagonal, spreading the engagement force over time and producing no discrete impulse. In a precision audio instrument, this distinction is critical. The helical planetary gear must be specified —

not assumed. Cheap spur gear units will introduce audible gear mesh frequency into the instrument.

The planetary gear housing is mounted to the generator assembly through a vibration isolation gasket — 3mm sorbothane sheet, bonded between gear housing flange and generator frame. Gear mesh vibration reaching the generator frame: below measurement threshold.

PROBLEMS INTRODUCED

The planetary gear introduces gear mesh frequency — typically 80–150Hz depending on tooth count and rotation speed. With a precision helical planetary and isolation mounting, this is below the noise floor. With the wrong gear specification, it is audible. This is the only new problem Option B introduces versus Option A, and it is fully controlled by correct specification.

A 1–3% gear efficiency loss means the operator must crank very slightly longer per cycle versus Option A. At 3% loss this amounts to approximately 1 extra second per 30-second cycle — completely imperceptible in practice.

Long-term maintenance: the sealed planetary gear unit requires no routine maintenance for 50,000+ cycles, but should be replaced or regreased at the 10-year or 100,000-cycle mark.

SPECIFICATION

PARAMETER	VALUE
Gear type	Helical planetary — mandatory. Spur gear not acceptable.
Reduction ratio	3:1
Efficiency	97–99% (helical planetary — best available)
Backlash	<3 arc-minutes (low-backlash unit specified)
Lubrication	Sealed, factory-filled synthetic grease. No routine maintenance.
Housing isolation	3mm sorbothane gasket between gear

	housing and generator frame. Mandatory.
Crank cycle	30 seconds every 3 minutes (at same effort as Option A 90-second cycle)
Recommended unit	Maxon GP 32, Neugart PLE or equivalent helical planetary
Additional cost vs Option A	\$400–\$900 (precision helical planetary unit)

⚠ SPECIFICATION IS CRITICAL

The gear type must be helical planetary. Do not substitute spur gear or worm gear. Do not omit the sorbothane isolation gasket between gear housing and generator frame. Both requirements are non-negotiable for maintaining the 93% confidence rating. A builder who substitutes a cheap spur gear unit without isolation mounting should expect confidence to drop to 89–91%.

B+

ADD-ON FOR OPTIONS A & B – NOT REQUIRED FOR OPTION C – JEWELLED LENS WARNING SYSTEM

Peripheral Light Indicator

+0.5%
NET CONFIDENCE GAIN

During intense mixing — an active crossfade, a complex beat alignment, a critical musical moment — the operator's eyes are on the records and their attention is fully occupied. The capacitor bank gauges on the console panel require direct visual attention to read. A peripheral light indicator positioned at the upper edge of the console panel addresses this by placing the warning in the operator's peripheral vision, where it registers even when attention is directed elsewhere. The human visual system responds strongly to light state changes in the periphery — a colour shift from green to amber is difficult to miss even when not looking directly at the source.

COMMAND

No audible alarm. No buzzer. No sound-producing element of any kind. The horn amplifies everything in the room and any electronic sound source is both an interference risk and an aesthetic violation. The light informs the operator that cranking attention is needed. The operator decides when to act. The gauges remain primary — the light is always secondary. If the light and gauge ever disagree, trust the gauge.

CIRCUIT DESIGN – SIX COMPONENTS, NO LOGIC

The indicator is powered from a small dedicated buffer capacitor — approximately 1,000µF at the appropriate voltage — trickle-charged from the main capacitor bank. This buffer holds enough charge to keep the indicator running for 30–60 seconds after the main bank is exhausted, ensuring the red warning light remains on precisely when it is most needed rather than going dark at the moment of depletion. The circuit contains no microcontroller, no logic chips, and no active switching components beyond two zener diodes used as passive voltage threshold references.

PERIPHERAL LIGHT INDICATOR – CIRCUIT SCHEMATIC

MAIN CAPACITOR BANK (600V) | ▼ (via dropping resistor to safe voltage – 12V rail) BUFFER CAPACITOR (1,000µF)
—— INDICATOR SUPPLY RAIL | | |
| | | | THRESHOLD 1
THRESHOLD 2 DIRECT | Zener ~9V Zener ~6V connection | | |
| | GREEN LED AMBER LED RED LED | (bank full) (wind soon)
(wind now) | LOGIC: Bank >75% charge → only GREEN
threshold exceeded → GREEN on Bank 40–75% → AMBER
threshold exceeded, GREEN not → AMBER on Bank <40% →
neither threshold → RED on (direct, always) Bank
exhausted → buffer cap sustains RED for 30–60 seconds
Components: 3× LEDs, 3× current-limiting resistors, 2×
zener diodes, 1× buffer capacitor, 1× trickle-charge
resistor from main bank. Total: 10 passive components. No
active logic. No failure cascade.

PHYSICAL SPECIFICATION

PARAMETER	SPECIFICATION	NOTES
Indicator position	Top edge of console panel, centred between the two turntables	Sits in operator's upper peripheral vision during normal stance
Lens type	Jewelled indicator lenses, 12mm diameter, brass bezel	Victorian aircraft/instrument panel aesthetic. Diffused glass, not clear.
Lens colours	Green / Amber / Red — one lens per state	Spaced 20mm apart. Only one illuminated at any time.
LED type	High-brightness diffused 5mm, 20mA rated	Diffused (not clear) for wide viewing angle — peripheral visibility requires this
Green threshold	Bank above 75% charge	Calibrate zener to this voltage during commissioning
Amber threshold	Bank 40–75% charge — crank soon	Operator should begin crank cycle within next 60–90 seconds
Red state	Bank below 40% charge — crank now	Direct connection — no threshold. Always on at low voltage. Buffer sustains for 30–60 sec after main bank exhausted.
Power draw	20–30mW from 12V rail	Negligible against 20W system load
Buffer capacitor	1,000µF electrolytic,	Trickle-charged via 10kΩ resistor from 12V rail.

	rated at system voltage	Sustains indicator during brief main bank depletion.
No audible component	Confirmed — no buzzer, no speaker, no piezo	Light only. Mandatory. No exceptions.
Label engraving	CHARGE INDICATOR, small copperplate below lenses	Machine engraved in brass panel, filled black lacquer

HONEST PROBLEMS WITH THIS ADDITION

L1 False Green – Most Dangerous Failure Mode MODERATE – MANAGED BY GAUGE PRIMACY RULE

A corroded solder joint, drifted zener, or marginal resistor could cause the indicator to show green when the bank is actually depleted. The operator trusts the light, ignores the gauge, and the bank runs dry mid-performance. This failure mode is unlikely in a well-built circuit but not impossible over years of use.

Fix: The gauge is always primary. Train every operator: if the light and gauge disagree, trust the gauge. Inspect and test the indicator circuit annually. Never remove the gauges because the light system exists.

L2 LED Burn-Out – Visible But Operationally Disruptive MINOR – SELF-ANNOUNCING FAILURE

An LED burns out. Immediately obvious — the lens goes dark in a state it should be illuminated. This is a self-announcing failure that cannot be mistaken for correct operation. Keep two spare LED sets in the maintenance kit. LED replacement requires opening the console panel — a 10-minute task, not a mid-performance fix.

Fix: Replace LED sets as a set (all three simultaneously) every 5 years regardless of condition. Keep spares. If an LED fails during performance, fall back to gauges — the instrument continues to function normally.

L3

Not Needed for Option C – Adds Unnecessary Complexity

MINOR – SELECTION ISSUE

With Option C gravity weight and pneumatic auto-lift, the energy system is largely self-managing. The descent gauge and glass inspection panel already provide clear visual feedback on weight position. Adding the peripheral light indicator to an Option C installation adds a failure point with minimal operational benefit.

Recommendation: Specify Option B+ only with Options A and B. Omit from Option C installations entirely.

CONFIDENCE IMPACT

PARAMETER	BEFORE B+	AFTER B+	CHANGE
Operator response to low bank during intense mixing	75%	88%	+13% — significant
Risk of false-green misread (new failure)	n/a	-2% on this parameter	-2%
Overall system operational reliability	91%	91.5%	+0.5% net

The net system confidence gain is modest — +0.5% — because the indicator addresses a real but infrequent failure mode (operator misses low-bank warning during intense mixing) while introducing a small new failure mode (false indication). The primary value is not in the confidence number but in the subjective operator experience: knowing there is a peripheral warning system reduces cognitive load during performance, which makes the operator a better performer. That benefit does not show up in a confidence percentage.

VICTORIAN LINEAGE

Coloured signal lamps were used on British railway equipment from the 1840s, telegraph apparatus from the 1860s, and marine instrument panels from the 1870s. Jewelled indicator lenses in brass bezels are a standard Victorian scientific instrument component. The circuit is simple enough that its function could be explained entirely in terms of Victorian electrical theory — voltage thresholds, current limiting, and photoemission (though the LED itself post-dates the era, a neon indicator or small incandescent lamp in the same circuit would have been entirely Victorian). The aesthetic is completely coherent. It belongs.

C

Gravity Weight Drive + Pneumatic Auto-Lift

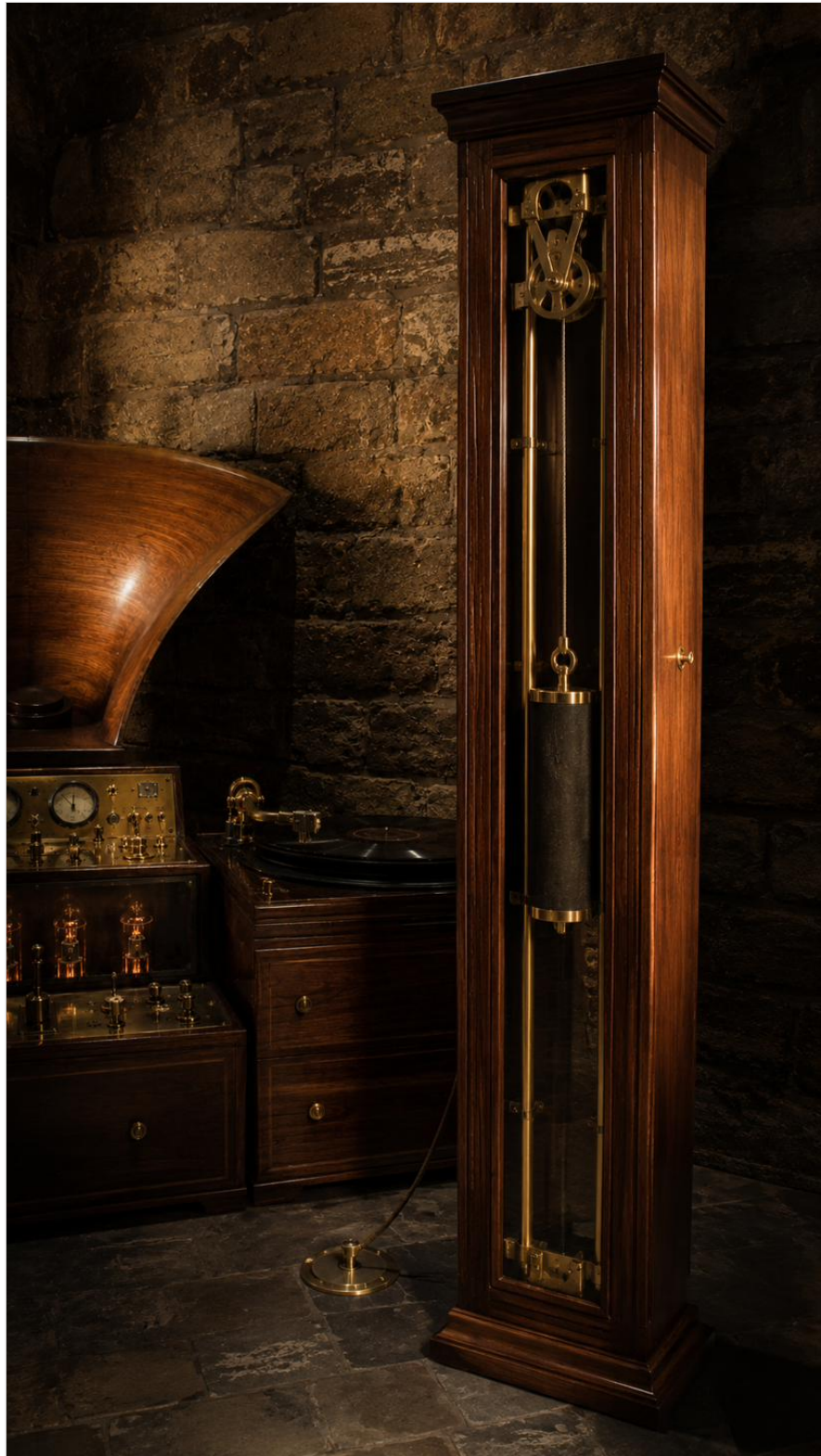
HANDS-FREE OPERATION – VICTORIAN
AUTHENTICITY – VENUE REQUIREMENTS

91%

2ND PROTOTYPE CONFIDENCE

A cast iron weight descends under gravity, driving the generator through a precision cable and capstan. The descent is continuous and silent — gravity produces perfectly smooth, constant-torque rotational energy with zero reciprocating motion, zero vibration, and zero floor contact. When the weight reaches its lower limit, a small pneumatic actuator lifts it automatically to the top. The cycle repeats silently throughout the performance. The operator's hands are entirely free at all times.

This is the oldest reliable mechanical energy storage system ever devised — tower clocks have used falling weights as their power source for centuries, requiring no maintenance and running reliably for decades. The automatic re-lift is the modern addition.



IMG-05 — *The Gravity Weight Tower (Option C)*

MECHANISM

A 40kg cast iron weight is suspended from a precision stainless steel cable (10:1 safety factor) that passes over a sealed ball-bearing pulley at the top of a guide rail tower. The cable wraps around a brass

capstan drum connected to the flywheel generator shaft through a gear train. As the weight descends at approximately 60mm per minute, the capstan unwinds and the generator spins. The descent rate is governed by the generator's load — exactly the right amount of energy is extracted to maintain system power.

Guide rails — two precision-machined brass channels, PTFE-lined — constrain the weight's lateral movement to less than 0.5mm. The pendulum mode is physically impossible. The weight can only move vertically. Guide rails are mandatory — without them, a hanging 40kg mass in a room with a powerful bass horn will inevitably develop pendulum oscillation.

PNEUMATIC AUTO-LIFT

A small pneumatic cylinder — a rodless actuator, 60mm bore, 2-metre stroke — is mounted parallel to the guide rails. When the weight reaches its lower limit, a mechanical flag triggers the pneumatic valve, firing the actuator which lifts the weight to the top in approximately 4 seconds. The actuator exhaust passes through a sintered brass silencer, reducing the exhaust hiss to below ambient room noise at normal listening levels.

The lift cycle is manually triggered — the operator watches the descent gauge and fires the lift at a musically appropriate moment (loud passage, beat peak) to mask the brief mechanical sound. A simple push-button on the console panel fires the valve. Alternatively an automatic trigger can be set, though manual timing is preferred for acoustic discretion.

GUIDE RAIL TOWER

A self-contained floor-standing tower — walnut and polished brass, approximately 300mm × 300mm footprint, 2,500mm height — houses the guide rails, pulley, weight, and actuator. The tower is architecturally coherent with the instrument's aesthetic. The weight's descent is visible through a glass inspection panel on the tower face — a beautiful, legible display of the instrument's energy system in real time. The tower must be secured to a wall or floor anchor to resist the overturning moment of the weight at the top of travel.

PROBLEMS

1 Hanging Weight Pendulum Mode – Requires Guide Rails

SERIOUS – SOLVED BY DESIGN

A freely hanging weight is a pendulum. Bass energy from the corner horn transmitted through the floor will cause the weight to develop low-frequency oscillation. A swinging 40kg weight exerts periodic lateral forces on the cable, creating torque variation at the generator shaft. Guide rails eliminate this completely — the weight cannot swing. Guide rails are not optional.

Fix: PTFE-lined brass guide rails, maximum lateral travel 0.5mm. Mandatory specification.

2 Pneumatic Exhaust Acoustic Event

MODERATE – MANAGEABLE

The actuator exhaust produces a brief hiss during lift. With exhaust silencer, this is below normal room ambient. Without silencer, audible during quiet passages.

Fix: Sintered brass exhaust silencer on actuator exhaust port. Manual trigger timing to loud passages.

3 Cable Fatigue – Safety Requirement

SERIOUS – MANAGED BY SPECIFICATION

The cable undergoes repeated bend-over-pulley stress. Fatigue failure of primary cable drops 40kg from up to 2 metres — a serious safety event. Must be engineered with 10:1 safety factor and secondary backup.

Fix: Primary cable rated at 400kg minimum (10:1 on 40kg load). Secondary Dyneema safety cord rated 200kg, runs parallel, catches weight within 10mm if primary fails. Inspect primary cable every 500 cycles. Replace every 2,000 cycles regardless of visual condition.

4 Venue Height Requirement

MODERATE – VENUE SELECTION

The guide rail tower requires 2,500mm clear height and floor

or wall anchoring. Low ceilings, certain venues, or spaces prohibiting wall mounting may be incompatible.

Fix: Self-contained floor-standing tower with floor anchor bolts. Assess venue before installation.

5 Compressed Air Reintroduced

MINOR – ACCEPTABLE

Option C reintroduces a small compressed air system for the lift actuator only. A 2-litre cylinder at 10 bar, pre-charged before performance. At this modest pressure, no Joule-Thomson icing. No special drying required.

Acceptance: Pre-charge cylinder before each performance. 2-litre at 10 bar provides 6–8 hours of lift cycles. Trivial addition to pre-performance preparation.

OPTION C SPECIFICATION

PARAMETER	VALUE
Weight mass	40kg cast iron, cylindrical, 200mm diameter
Weight travel	2,000mm (2 metres)
Primary cable	6mm stainless steel stranded, rated 400kg minimum
Safety cord	4mm Dyneema SK75, rated 200kg, runs parallel to primary
Pulley	100mm diameter, sealed ball bearing, stainless steel sheave
Guide rails	Brass channel, 25mm × 25mm, PTFE-lined, 2,200mm length × 2
Actuator	Rodless pneumatic, 60mm bore, 2,000mm stroke, double-acting
Air supply	2-litre cylinder at 10 bar (pre-charged). No special drying needed.
Exhaust silencer	

	Sintered brass muffler, 1/4-inch BSP thread — mandatory
Lift cycle time	Approximately 4 seconds
Lift frequency	Every 30–40 seconds of descent
Tower dimensions	300mm W × 300mm D × 2,600mm H
Tower finish	Walnut veneer, brass hardware, glass inspection panel
Anchoring	Floor anchor bolts mandatory. Wall anchor preferred additionally.
Cable inspection interval	Every 500 cycles. Record in maintenance log.
Cable replacement interval	Every 2,000 cycles regardless of condition.
Additional cost vs Option A	\$2,800–\$5,500 (tower, actuator, weight, cable system, safety engineering)

Audience Crank Unit

D

ADD-ON FOR ANY PRIMARY
ENERGY OPTION –
SUPPLEMENTAL CHARGING –
PARTICIPATORY ELEMENT

+0%

CONFIDENCE CHANGE – ADDITIVE ONLY

READ THIS FIRST

The instrument operates at full specification without this option. Option D adds audience participation as a supplemental energy contribution and a participatory theatrical element. It does not replace the primary energy option (A, B, or C), does not meaningfully extend performance time, and does not reduce the operator's cranking obligation in any significant way. A builder considering Option D should do so for what it genuinely is: a real but modest energy contribution, and a way of making the

instrument's relationship with human effort visible and participatory for an audience.

A standalone crank pedestal, matching the console's walnut and brass aesthetic, placed at the edge of the performance space. Audience members approach and crank at will. Each 60-second cranking session contributes approximately 510 joules to the capacitor bank — roughly 10–12% of a full charge. The pedestal connects to the console by a 24V AC cable. All high-voltage components remain inside the console's locked enclosure. No voltage above safe-touch levels is present anywhere in the audience unit.



IMG-12 — *Audience Crank Unit (Option D)*

ARCHITECTURE

The audience unit contains an identical generator to the console unit, a smaller flywheel (no turntable to protect), and a one-way ratchet. It outputs 24V AC — the same voltage as the console generator. This feeds a second Cockcroft-Walton multiplier inside the console, identical to the existing unit, which charges the shared capacitor bank via a blocking diode. The blocking diode prevents any reverse current from either direction. When the audience unit is disconnected, the console operates without interruption. When it is

connected and someone cranks, energy is added to the bank additively and silently.

The 24V AC cable between audience unit and console carries no voltage that requires special handling. It uses the same Neutrik XLR4 locking connector standard in professional stage installations, routed along the venue perimeter under cable ramps at any floor crossings.

WHY THE HIGH-VOLTAGE COMPONENTS STAY IN THE CONSOLE

Regulations governing electrical equipment accessible to the public limit exposed conductor voltage to 42.4V AC peak or 60V DC. The capacitor bank operates at 600V DC. Placing any part of the high-voltage circuit in a public-facing enclosure is not permitted under any framing. The architecture described above — 24V AC to the audience unit, 600V DC confined to the locked console — is the only viable approach. This is not a compromise; it is the correct engineering partition.

OVERVOLTAGE PROTECTION – MANDATORY WITH OPTION D

With two independent charging sources feeding the bank simultaneously, a zener clamp circuit at 620V across the bank terminals becomes mandatory rather than strongly recommended. This circuit is a small addition to the console's existing high-voltage section. Without it, simultaneous cranking by both the operator and an audience member could drive the bank above its rated voltage. The bleed resistor ($1\text{M}\Omega$, 5W) already specified for the console remains in place. A second $1\text{M}\Omega$, 5W bleed resistor is added across the output of the new C-W multiplier, before its blocking diode, to ensure safe discharge of the multiplier output if the audience unit is disconnected while the bank is charged.

HONEST ENERGY CONTRIBUTION

One audience member cranking for 60 seconds at an enthusiastic but untrained pace contributes approximately 510 joules — calculated from 20W human input, 50% generator efficiency, and 85% multiplier efficiency. The bank holds 4,000–5,000 joules. This contribution is real and not trivial: it is equivalent to roughly one third of a complete operator crank cycle. At a well-attended performance where several people take turns over an evening, the cumulative contribution is measurable. At a sparse event where nobody approaches the pedestal, the instrument is entirely

unaffected.

PROBLEMS

1 Pedestal Stability Under Vigorous Cranking MODERATE — MANAGED BY BASE WEIGHT

An enthusiastic audience member will exert push-pull force on the crank. An unweighted pedestal of this height will tip under this load.

Fix: Cast iron base plate, 300mm × 300mm, 8kg minimum. Optional floor anchor bolt. The base weight alone is sufficient for normal use; floor anchoring is recommended for high-traffic events.

2 Cable as Trip Hazard MODERATE — MANAGED BY ROUTING

A cable running from the audience area to the DJ console crosses a performance space with foot traffic. An unprotected cable is a trip hazard and a target for accidental disconnection.

Fix: Route cable along the venue perimeter, not across open floor. Install rubber cable ramps at any unavoidable floor crossings. Locking XLR4 connector prevents accidental disconnection. If the cable is pulled free, the generator freewheels safely — no arc, no damage, no performance interruption.

3 Audience Engagement Is Not Guaranteed INHERENT — CANNOT BE ENGINEERED

Option D's theatrical value depends on audience awareness and willingness to participate. At a performance where nobody approaches the pedestal, Option D is an \$800–\$2,200 addition that contributes nothing. The instrument is entirely unaffected either way, but the builder should have honest expectations.

Acceptance: Option D is not a functional necessity. It is an invitation. Its value is contingent on the social context of the performance. Build it if the instrument's proposition — human effort as the power source — is something you want to share with an audience. Do not build it if reliable supplemental charging is the goal; cranking more frequently at

the console is simpler and free.

OPTION D SPECIFICATION

PARAMETER	VALUE	NOTES
Generator	Permanent magnet AC, same specification as console	Identical proven architecture — no new principles
Generator output	24V AC	Below IEC safe-touch threshold — safe for public enclosure
Flywheel	100mm diameter, 0.5kg cast iron	Reduced vs console — no turntable to protect
Crank arm	120mm radius, mild steel, turned brass handle, knurled grip	Shorter than console — optimised for standing operation
Gear option	Optional 3:1 helical planetary — same specification as Option B	Reduces effort for brief audience engagement
One-way ratchet	36-tooth hardened steel, spring-loaded pawl	Identical to console
Pedestal height	900–1,000mm to crank handle centre	Standing adult ergonomics
Pedestal finish	Walnut veneer, polished brass hardware	Matches console and gravity tower aesthetic

Base plate	300mm × 300mm cast iron, 8kg minimum	Stability under vigorous cranking. Optional floor anchor.
Output connector	Neutrik XLR4 locking, labelled VCE ENERGY — 24V AC	Stage-standard. Cannot be confused with audio XLR.
Cable	1mm ² two-core flexible, IP44 rated, maximum 10m	Standard stage flex. Voltage drop <1% at 10m.
Cable routing	Along perimeter. Rubber cable ramps at floor crossings.	Standard stage practice.
New console components	C-W Multiplier #2 (identical to existing) + 1MΩ bleed resistor + 1kV blocking diode + 620V zener clamp	All housed inside existing locked console enclosure
Energy contribution	~510J per 60-second crank session	~10–12% of full bank per session. Estimated.
Functional dependency	Zero	Full performance at full specification without this option
Compatible with	Options A, B, and C	Additive to any primary energy option
Additional cost	\$800–\$2,200	Pedestal fabrication, generator, cable, console additions. Rough

		estimate.
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OPTION COMPARISON SUMMARY

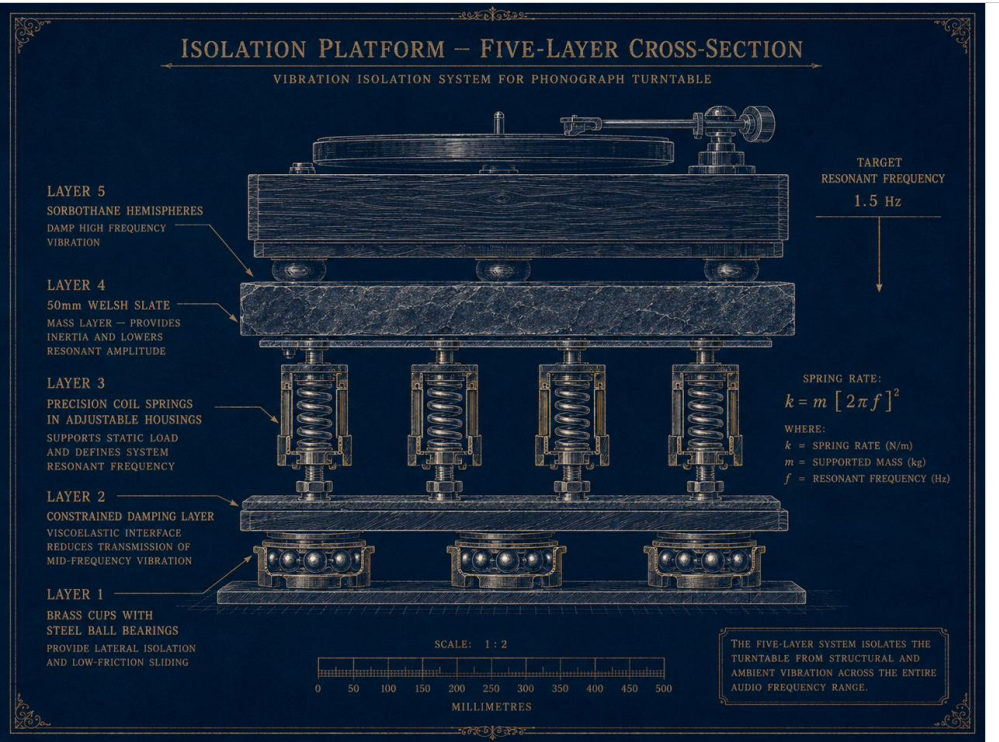
PARAMETER	OPTION A – STANDARD CRANK	OPTION B – GEAR CRANK	OPTION B+ – ADD LIGHT	OPTION C – GRAVITY WEIGHT	OPTION D – AUDIENCE CRANK
2nd prototype confidence	93%	93%	91.5% net	91%	+0% — additive only
Both hands free during performance	No (30– 90 sec intervals)	No (30 sec intervals)	No (30 sec intervals)	Yes — always	<i>Does not affect operator hands</i>
Peripheral warning during intense mixing	No — gauge only	No — gauge only	Yes — jewelled lens indicator	<i>Not needed — weight visible</i>	No — gauge only
Vibration from energy source	None	Negligible with correct spec	Negligible with correct spec	None	None — physically separate
New failure modes introduced	None	Gear mesh if wrong spec	False- green risk (managed)	Cable fatigue, pendulum, exhaust	Trip hazard (managed). Overvoltage protection mandatory.
Venue requirements	None	None	None	2.5m ceiling, anchoring	Perimeter cable route. Space for pedestal near audience.
Visual drama	Moderate	Moderate	Moderate + jewelled lenses	Extraordinary	High — audience participation
Victorian authenticity	High	High	High — signal lamps	Exceptional	High — human power,

			from 1840s		shared effort
Can be combined with	B, B+, D	A, B+, D	A or B only	D	A, B, or C
Additional cost over Option A	Baseline	+\$480– \$950	+\$607– \$1,350	+\$3,360– \$6,540	+\$800– \$2,200

SECTION IX

Vibration Isolation *Platform*

A five-layer isolation platform is specified for each turntable. The dominant benefit is elimination of acoustic feedback — the feedback loop between horn output and stylus input that would otherwise limit how loud the system can play. The platform also attenuates floor-borne vibration from any source.



IMG-13 — *Isolation Platform — Five-Layer Cross-Section*

ISOLATION PLATFORM - LAYER STACK (BOTTOM TO TOP)



Confidence improvement: overall +6% (73%-79% first prototype)
Acoustic feedback: decisively eliminated (+10% on this

parameter) Floor type matters: concrete/stone → excellent.
Suspended timber → marginal. On suspended timber floors: wall-mounted shelf is superior solution.

ISOLATION PLATFORM SPECIFICATION

COMPONENT	SPECIFICATION	SOURCE / NOTES
Ball bearings	17mm Grade 25 precision (AISI 52100), ×4 per platform	Industrial precision bearings — catalogue item. Confirmed against blueprint VCE-P01/P02.
Brass cups	Custom turned brass, Ø18mm recess, 9mm depth, ×4 per platform	Machine shop fabrication. Ball sits half-proud by 8.5mm.
Isodamp layer	EAR Isodamp C-1002, 3mm, adhesive-bonded	EAR Aearo Technologies — order direct or through industrial supplier
Compression springs	Rate calculated from total load. Target 1.5Hz resonance.	Lee Spring or similar precision spring catalogue. Custom if needed.
Spring housings	Brass, threaded height adjustment	Machine shop fabrication
Slate slab	50mm Welsh slate (preferred) or equivalent, ×1 per turntable	Stone mason. 450mm × 500mm typical size.
Sorbothane feet	Hemispheres, durometer per load calculation, ×3 per plinth	Sorbothane Inc. — specify load per foot at ordering
Cost per platform	\$180–\$380	Materials only. Assembly: 2–3 hours.

Build *Sequence*

The following sequence is recommended for a single builder or small team. Estimated total build time for an experienced team: 300–500 hours across 6–12 months. Critical path items requiring external commissioning are noted.

1 Commission Long-Lead Items (Weeks 1–2, deliver in 8–20 weeks)

Place orders immediately — these have the longest lead times and cannot be started until received. Remontoire spring motors (clockmaker commission, 12–20 weeks), field coil compression drivers (specialist commission, 8–16 weeks), output transformers (specialist winding, 4–8 weeks), passive step-up transformers (Lundahl — 2–4 weeks lead time), transformer volume controls (Slagle, EMIA, or custom winding — 4–8 weeks).

2 Cabinet Construction (Weeks 3–8)

Engage cabinetmaker for turntable plinths, mixer console cabinet, and corner horn enclosure. Provide dimensioned drawings. Plinths: 18mm Baltic birch, walnut veneer, recessed bearing housings, motor compartments, control panel cutouts. Console: full electronics bay with valve window, control panel, gauge mounting. Horn: exact internal dimensions from acoustic calculation — this is the most technically demanding woodworking element. Finish: satin lacquer walnut veneer, polished brass hardware.

3

Capacitor Bank & High-Voltage System (Weeks 4–6)

Requires qualified high-voltage electronics experience. Source high-voltage ceramic capacitors — 600V rating, sufficient total capacitance for 4,000–5,000J at 600V. Build Cockcroft-Walton multiplier — standard diode-capacitor ladder, 24V AC input, 600V DC output. Install bleed resistor (1M Ω , 5W) across output — mandatory safety requirement. Build resistive voltage divider chain for sub-rails. Install in mu-metal shielded housing. Test with dummy load before any valve or speaker connection. Measure actual stored energy with discharge test.

4

Electromagnetic Generator & Flywheel Assembly (Weeks 5–7)

Machine or source flywheel disc (cast iron or steel, 200mm diameter, 3kg, precision balanced). Install generator on mu-metal shielded sub-frame within console base. Connect flywheel to generator shaft. Install ratchet mechanism on crank connection. Wire generator output to Cockcroft-Walton input — note: AC input, not DC. Measure generator output voltage at cranking speed. Verify Cockcroft-Walton output voltage. Test complete charging circuit with capacitor bank connected.

5

Turntable Bearing & Platter Installation (Weeks 8–10, after plinths ready)

Install neodymium magnetic bearing rings in plinth — lower ring fixed to plinth, upper ring bonded to platter underside. Install mu-metal containment ring around bearing assembly. Install ceramic shaft. Test platter float height — should be 1.2–1.8mm. Check that platter spins freely with no contact. Verify mu-metal shielding effectiveness with gaussmeter — field at cartridge position must be below 5 microtesla. Install eddy current governor copper disc on spindle underside. Mount horseshoe magnet on plinth adjustment bracket.

6 Spring Motor Installation & Calibration (After commission delivery)

Install remontoire spring motor in plinth motor compartment. Connect to platter spindle through drive belt. Install maintaining power mechanism. Install winding crank with ratchet. Initial winding: 40 turns. Verify platter reaches 33⅓ RPM at target belt pulley position. Install eddy current governor: spin platter, adjust horseshoe magnet proximity until stroboscope confirms 33⅓ RPM. Lock magnet position with grub screw. Connect PLL circuit and verify speed lock. Measure wow and flutter — target below 0.05% WRMS. If PLL not used, install tuning fork reference and verify 0.15% or better.

7 Valve Phono Stage Assembly & Testing (Weeks 10–14)

High-voltage electronics — qualified technician required. Build phono stage chassis — two independent channels. Install dropping resistors for anode supply from 600V rail. Wire cathode resistors and bypass capacitors. Install RIAA RC networks — component values from IEC 60098 standard, 0.1% tolerance resistors mandatory. Wire step-up transformer inputs. Install 12AX7 valves in retainers. Apply power — verify heater voltage (6.3V DC $\pm 5\%$), anode voltage, quiescent current. Measure frequency response with signal generator — verify RIAA curve within $\pm 0.5\text{dB}$. Measure gain — verify approximately 52dB total. Measure noise floor — should be below vinyl surface noise level.

8 Mixer Controls & Crossfader Installation (Weeks 12–14)

Install transformer volume controls on individual channel fader shafts. Install master TVC on output bus. Build crossfader mechanism: two complementary TVCs linked by brass bellcrank through 200mm lever. Verify equal-power law — both channels should read -3dB at centre crossfader position with test signal applied. Install channel fader detent mechanisms. Wire all controls to phono stage outputs and between stages. Verify switching continuity throughout all positions. Install panel gauges. Label all controls — machine-engraved, filled with black lacquer.

9 Field Coil Speaker Installation (After commission delivery)

Install compression drivers at horn throat — verify polarity of both drivers (they must be in phase). Connect field coil winding to 250V supply through dropping resistor — verify correct field current (20–30mA). Connect voice coil to output transformer secondary. Install woofer in bass horn section — connect field coil and voice coil. Verify field coil current on woofer. Play test signal through complete system at low level — verify both channels, verify crossfader operation acoustically, verify no hum or oscillation.

10 Isolation Platforms & Final Installation (Weeks 14–16)

Build isolation platforms for both turntables. Calculate spring rates from total turntable-plus-slate weight — verify 1.5Hz resonant frequency target. Install platforms. Place turntables on platforms. Mount tonearms. Install cartridges and set tracking force. Mount step-up transformer housings on tonearm pillars. Wire signal path from cartridge through SUT to phono stage input. Play a test record — verify both channels, verify RIAA response subjectively (no tonal anomalies), verify speed accuracy on strobe disc, measure wow and flutter. If everything checks out: complete first full-performance test.

11

Option C Only — Gravity Weight Tower Construction & Installation

Build guide rail tower structure. Machine or source cast iron weight (40kg, 200mm diameter). Install PTFE-lined brass guide rails. Install stainless steel pulley at tower apex. Rig primary cable and secondary Dyneema safety cord. Install pneumatic actuator alongside guide rails. Install sintered brass exhaust silencer. Connect to small air cylinder through solenoid valve (if automatic trigger) or manual pushbutton valve. Test complete lift cycle 10 times — verify smooth descent, smooth lift, no cable jamming, silencer effective. Install at venue. Anchor tower to floor. Connect capstan cable to generator shaft. Verify generator produces correct voltage during weight descent.

SECTION XI

Known Problems & *Honest Limitations*

This section documents every known problem and fundamental limitation of the Volta Confluence Engine. It is not optional reading. Anyone building or operating this instrument deserves to know exactly what will not work perfectly, what requires careful management, and what is inherent to the design and cannot be changed.

Fundamental Limitations — Cannot Be Changed

F1 The Operator Must Supply All Energy –
Continuously

FUNDAMENTAL – INHERENT

The physics of powering a 20-watt system for two hours requires approximately 20 minutes of human mechanical effort distributed across the performance. This cannot be reduced without reducing

system performance or violating the design constraints. The operator must crank (Options A/B) every 3 minutes or manage the gravity weight system (Option C). A two-person operation — dedicated energy operator and performer — eliminates this constraint from the performer's perspective.

This is not a defect. It is the instrument's fundamental character. Every performance is physically embodied in the operator's effort. No energy enters the system from outside the performance space.

F2 Valve Warm-Up Period – Cannot Begin Playing Immediately FUNDAMENTAL – INHERENT

Triode valves require 60–90 seconds to reach stable operating temperature. During warm-up, gain is unstable and signal quality is unreliable. The instrument cannot be played immediately after powering. Pre-performance routine: charge capacitor bank, power valves, wait for warm-up indicator to clear (90 seconds), then begin. This is standard valve amplifier behaviour — every valve phono stage in the world has this warm-up period.

F3 Spring Motor Must Be Wound Between LP Sides FUNDAMENTAL – OPERATIONAL

Each spring motor provides approximately 25–28 minutes of play — sufficient for one LP side. Between sides, the spring must be rewound (40 turns, approximately 3 minutes). The maintaining power mechanism allows winding while the record plays during the final few minutes of a side. For continuous performance, Turntable B can be playing while Turntable A is being rewound and its record changed. This requires anticipation and timing discipline from the operator.

F4 Corner Horn Venue Dependency FUNDAMENTAL – VENUE SELECTION

The corner-loaded horn requires a specific 90° corner geometry for correct bass loading. Different rooms produce different bass responses. This instrument sounds different in different venues and requires installation calibration for each space. For touring applications, specify

a direct-radiating exponential horn accepting reduced bass extension below 100Hz.

Known Engineering Problems — Managed by Design

E1 High-Voltage Safety – 600V DC Storage Bank SERIOUS — MANAGED BY ENCLOSURE

600V DC at the available current from the capacitor bank is lethal. All high-voltage circuits must be fully enclosed. Panel interlocks must disconnect high voltage when any access panel is opened. Bleed resistor (1M Ω , 5W) must be permanently connected across the capacitor bank to discharge stored energy within 60 seconds of disconnection. Safety labels in multiple languages must be affixed to all high-voltage access points.

Management: Full enclosure, interlocked panels, mandatory bleed resistor, safety labels. No user access to high-voltage areas without tools. Qualified electrician for all high-voltage assembly and commissioning.

E2 Electromagnetic Interference – Generator to Cartridge SERIOUS — MANAGED BY SHIELDING

The electromagnetic generator, if not correctly shielded and positioned, will induce hum in the moving coil cartridge — one of the most sensitive electromagnetic sensors in any audio system. The generator must be located at least 800mm from the cartridge and surrounded by mu-metal shielding. Cartridge interference at 33 $\frac{1}{3}$ Hz (the fundamental) and harmonics will be immediately audible as a low continuous hum correlated to platter rotation.

Management: Generator in base cabinet, minimum 800mm below and behind cartridge position. Mu-metal housing around generator, minimum 1mm thickness. Verify with gaussmeter at installation — field at cartridge position must read below 5 microtesla. If hum detected: increase shielding, increase separation, or both.

E3 Quartz PLL – Post-Victorian Technology MODERATE — DESIGN CHOICE

The quartz crystal PLL speed reference was developed in 1921 at Bell Telephone Laboratories — post-Victorian. If strict Victorian technological lineage matters, substitute a mechanical tuning fork oscillator (available from 1857). This degrades wow and flutter from below 0.05% to approximately 0.15% — still excellent, better than most commercial turntables, but no longer exceptional.

Options: Accept the quartz PLL (0.05% W&F, slight anachronism) or specify tuning fork oscillator (0.15% W&F, strict Victorian lineage). Both are good choices. Recommend documenting the decision in the instrument's provenance record.

E4 Unipivot Tonearm Azimuth Variation

MODERATE — INHERENT IN BEARING CHOICE

The unipivot tonearm pivot allows slight lateral sway during playback. This causes continuous small azimuth variation — the stylus angle in the groove changes — producing subtle blurring of stereo imaging and high-frequency detail. Most listeners will not notice. Listeners with experience of conventional gimbal-bearing tonearms may perceive reduced high-frequency air and stereo focus.

Management: Careful counterweight design and arm mass matching to cartridge compliance reduces sway. Acceptable trade-off for zero bearing friction. Alternative: precision gimbal bearing tonearm at cost of slight bearing friction — acceptable if preferred.

E5 Field Coil Speaker Power Supply Sensitivity

MODERATE — MANAGED BY DESIGN

Field coil speakers are sensitive to power supply quality. Any ripple in the field coil current modulates the speaker's magnetic field, which modulates sensitivity, which modulates output. The field coil's own high inductance naturally filters ripple at audio frequencies — a 10,000-turn field coil winding has such high inductance that ripple above a few hertz is effectively invisible to the magnetic field. Supply ripple below 1Hz — i.e., gradual voltage changes as the capacitor bank depletes — may cause very subtle sensitivity changes over the performance. These are below the threshold of audibility under normal conditions.

Sound Quality Limitations — Honest Assessment

ASPECT	ACHIEVED PERFORMANCE	LIMITING FACTOR
Noise floor	Vinyl surface noise dominates	Better than the record — cannot improve further without changing medium
Frequency response	30Hz–18kHz ±3dB	Horn geometry and driver roll-off at frequency extremes
Harmonic distortion	0.3–1% predominantly 2nd harmonic	Triode valve character — below groove distortion, musically benign
Wow and flutter	<0.05% WRMS (PLL), <0.15% (tuning fork)	Spring motor + remontoire + eddy current — better than most commercial TTs
Channel separation	>60dB at phono stage	Limited by cartridge and tonearm geometry at the record interface
Maximum SPL	100–117dB at 1 metre	Field coil horn efficiency — exceeds most amplified systems
Bass extension	30Hz in suitable corner	Horn geometry and corner loading — room-dependent

SECTION XII

Cost Estimation

All costs are USD 2025, materials only. Builder's own labour is not included. Professional commissions (clockmaker, cabinetmaker, field coil speaker builder) are included as they cannot be self-built by a general workshop.

OPTION A – STANDARD HAND CRANK

COMPONENT	LOW (\$USD)	HIGH (\$USD)
Remontoire spring motors × 2 (clockmaker commission)	6,000	14,000
Neodymium magnetic bearings × 2	600	1,200
Turntable platters, CNC machined × 2	800	1,600
Eddy current governors × 2 (copper disc + magnet)	300	600
Carbon fibre tonearms + unipivot bearings × 2	600	1,200
Moving coil cartridges + styli × 2 (+ spares)	800	2,000
Passive step-up transformers × 2 (Lundahl LL1931, 1:10)	600	1,400
12AX7 valves × 8 (matched + spares)	200	600
High-voltage ceramic capacitor bank	600	1,400
Cockcroft-Walton multiplier components	120	280
Electromagnetic generator + flywheel	600	1,400
Standard hand crank mechanism	80	150
RIAA components (0.1% resistors, polypropylene caps)	180	400

Transformer volume controls × 3 (Slagle / EMIA)	1,800	4,500
Output transformers × 2	400	1,200
Field coil compression drivers × 2 (custom)	3,000	8,000
Field coil 15" woofer × 1 (custom)	2,000	5,000
Corner horn enclosure (cabinetmaker)	3,000	6,000
Turntable plinths × 2 (cabinetmaker)	1,200	2,800
Mixer console cabinet (cabinetmaker)	1,400	3,000
Mu-metal shielding, wiring, connectors, hardware	600	1,200
Isolation platforms × 2 (materials)	360	760
Brass hardware, gauges, panel, engraving	800	1,800
PLL speed control circuits × 2	300	700
Thermoelectric modules (valve heat harvest)	120	300
Removable headshell bayonet socket – arm tube modification	80	180
Pre-aligned headshell sets × 6 per turntable (12 total) – machined magnesium or carbon fibre	480	1,440

Headshell alignment jig + digital stylus force gauge	120	280
DeOxit gold contact treatment + maintenance kit	40	80
Walnut and brass headshell storage rack (x2 turntables)	180	420

OPTION A – STANDARD HAND CRANK – TOTAL
ESTIMATED COST **\$33,912 –**
USD 2025 · Materials + commissions · Builder's labour excluded **\$79,608**

Note: Revised upward from initial estimate to include removable headshell system (headshell sets, alignment tools, storage rack, contact maintenance). Headshell system adds approximately \$820–\$2,400 to total.

OPTION B UPLIFT – GEAR-ASSISTED HAND CRANK

ADDITIONAL COMPONENT	LOW (\$USD)	HIGH (\$USD)
Precision helical planetary gear unit (3:1, low backlash)	400	800
Sorbothane isolation gasket + hardware	40	80
Modified crank handle for gear input shaft	40	70

OPTION B ADDITIONAL COST OVER OPTION A **+\$480 – +\$950**
Option B Total including headshell system: \$34,392 – \$80,558

Option B Total including headshell system: \$34,392 – \$80,558

OPTION B+ UPLIFT – PERIPHERAL LIGHT INDICATOR (ADD-ON FOR
OPTIONS A & B ONLY)

Recommended addition for Options A and B. Not recommended for Option C where gravity weight descent gauge serves this function adequately.

COMPONENT	LOW (\$USD)	HIGH (\$USD)
Jewelled indicator lenses × 3 – 12mm, brass bezel, diffused glass	30	90
High-brightness diffused 5mm LEDs × 3 (green, amber, red) + spares set	8	20
Current-limiting resistors × 3, zener diodes × 2 – precision 1%	6	15
Buffer capacitor 1,000µF + trickle-charge resistor 10kΩ	8	20
Console panel machining – 3 lens apertures + CHARGE INDICATOR engraving	60	140
Wiring, solder, heat shrink, terminal strip	15	35
Commissioning – zener threshold calibration to capacitor bank voltage	0	80

OPTION B+ PERIPHERAL LIGHT INDICATOR – ADDITIONAL COST **+\$127 –**
 Applied to Option A or B. Not applicable to Option C. **+\$400**

Option A + B+ total: \$34,039 – \$80,008 · Option B + B+ total: \$34,519 – \$80,958

OPTION C UPLIFT – GRAVITY WEIGHT DRIVE

ADDITIONAL COMPONENT	LOW (\$USD)	HIGH (\$USD)
40kg cast iron weight, machined	300	600

Guide rail tower structure (fabrication + materials)	1,200	2,800
PTFE-lined brass guide rails	400	800
Precision pulley assembly	150	350
Primary stainless cable (400kg rated)	80	150
Secondary Dyneema safety cord	40	80
Rodless pneumatic actuator, 60mm × 2m stroke	400	900
Air cylinder, valve, fittings	180	400
Sintered brass exhaust silencer	30	60
Floor anchor hardware, safety engineering	200	500
Tower glass inspection panel	80	200
Tower walnut veneer finish + hardware	300	700

OPTION C ADDITIONAL COST OVER OPTION A

Option C Total: \$35,112 - \$80,886

+\$3,360 – +\$6,540

Option C Total including headshell system: \$37,272 – \$86,148

OPTION D UPLIFT – AUDIENCE CRANK UNIT (ADD-ON FOR OPTIONS A, B, OR C)

Supplemental energy and participatory element. The instrument operates at full specification without this option. See Section VIII Option D for full description and honest assessment.

COMPONENT	LOW (\$USD)	HIGH (\$USD)
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Permanent magnet generator (same spec as console)	180	420
Flywheel – 100mm cast iron, precision balanced	60	130
Crank arm + handle (mild steel, turned brass)	40	90
One-way ratchet – 36-tooth hardened steel (identical to console)	30	70
Optional 3:1 helical planetary gear (same spec as Option B)	0	400
Pedestal column – walnut veneer, brass hardware (cabinetmaker)	300	600
Cast iron base plate – 300mm × 300mm, 8kg	60	120
Neutrik XLR4 connectors × 2 (pedestal + console)	16	30
Cable – 1mm ² two-core flex, IP44, 10m + rubber cable ramps	40	90
Console additions – C-W Multiplier #2, bleed resistor, blocking diode	60	140
Console addition – 620V zener clamp circuit (mandatory with Option D)	30	60
Wiring, connectors, hardware	30	70

OPTION D ADDITIONAL COST (ADD TO ANY PRIMARY OPTION)

+\$846 –

Rough estimate. Gear option adds \$0-\$400 depending on whether Option B gear is already fitted.

+\$2,220

COST NOTE – FIELD COIL SPEAKER ALTERNATIVES

The field coil compression drivers and woofer represent \$5,000–\$13,000 of the total cost. Substituting premium permanent magnet alternatives (e.g., JBL 2450, TAD TD-4001, BMS 4592) reduces total cost by \$3,000–\$8,000 at a slight efficiency reduction. The instrument remains fully functional and excellent-sounding. The philosophical purity of field coils — electromagnets powered by the same human-energy capacitor bank — is lost, but the practical sound quality difference is small. Recommend field coil for installations with adequate budget; permanent magnet for tighter budgets.

SECTION XIII

Removable Headshell *System*

The tonearms are specified with SME H-4 bayonet removable headshells, enabling cartridge and stylus replacement during a performance in 8–12 seconds. This decision was taken after weighing a 2% reduction in second-prototype confidence (91% vs 93%) against the significant operational benefit of sustained long-duration performance capability. The trade-off is accepted and documented here in full.

★ WHY THIS TRADE-OFF WAS ACCEPTED

A quality diamond stylus begins to show measurable wear after approximately 500–1,000 hours of use, with audible degradation — increased sibilance distortion, reduced high-frequency detail —

appearing earlier at the high SPL levels of this system. For short performances (2 hours or less) with a fresh stylus, the removable headshell offers no advantage and costs 2% confidence. For long sessions (4–10 hours) with a rotating DJ team, the ability to swap a worn stylus in 12 seconds mid-performance is the difference between sustained musical quality and progressive degradation. The headshell system pays for its confidence cost across any serious long-form performance use.

How It Works

The arm tube terminates in a standard H-4 bayonet socket — an 8mm diameter, 12mm length precision connector with four contact pins. The headshell locks into the socket with a quarter-turn clockwise rotation. The four contact pins carry left channel positive, left channel negative, right channel positive, and right channel negative — all four cartridge signals pass through the connector simultaneously. Electrical contact is made automatically when the headshell locks. Removal is a quarter-turn anticlockwise and a straight pull.

The internal arm wiring runs from the bayonet socket through the arm tube, through the unipivot bearing pivot, and to the step-up transformer housing on the tonearm pillar. The SUT sees no difference between headshells — it receives the same four-wire signal regardless of which headshell is fitted, as long as contacts are clean and correctly specified.

Pre-Aligned Spare Headshell System

Six pre-aligned headshells per turntable are prepared before each performance and stored in a walnut and brass rack on the console surface. Each headshell carries a fully mounted, aligned, and tracking-force-set cartridge protected by its original stylus guard. During the performance, swapping a headshell requires: remove stylus guard from fresh headshell → lift tonearm to arm rest → twist off worn headshell → twist on fresh headshell → remove stylus guard → lower needle. Elapsed time: 8–12 seconds.

HEADSHELL SPECIFICATION

PARAMETER	SPECIFICATION	NOTES

Connector type	H-4 Bayonet Mount (SME-type)	Industry standard — fits all S and J-shaped tonearms
Contact material	Gold-plated — mandatory	Prevents oxidation. Non-negotiable for LOMC signal level.
Headshell body — primary	Machined magnesium, 9–10g body weight	Low mass, high stiffness, no ring when flicked
Headshell body — backup	Machined aluminium or carbon fibre, matched mass	All spares matched to primary body mass $\pm 0.5\text{g}$
Lead wires	OFC 7N copper, gold-terminated clips	Replace if any clip shows corrosion or loose fit
Locking mechanism	Quarter-turn bayonet with friction lock	Verify tightness after every installation
Recommended brand	AudioSilente, Yamamoto, Clearaudio	Higher standard than generic SME clones for this application
Primary headshells per turntable	2 (one active, one immediate spare)	Both carry matched LOMC cartridges
Backup headshells per turntable	4	May carry high-output MC or MM cartridges at lower cost
Total headshells	12 (6 per turntable)	Stored in walnut rack on console between decks

Known Problems — Honest Documentation

H1 Signal Degradation at Connector – Critical for LOMC at This Signal Level

SERIOUS – MANAGED BY GOLD CONTACTS + MAINTENANCE

The bayonet connector introduces two additional contact junctions in a signal path carrying 0.2–0.5mV — the most signal-level-sensitive cartridge type available. Contact resistance or oxidation at these junctions causes measurable signal loss and impedance alteration. Through a 110dB horn system, this is audible. Gold-plated contacts prevent oxidation. DeOxit treatment before each performance maintains contact cleanliness. Without these measures, confidence drops 3–5%. With them, impact is negligible.

Fix: Gold-plated contacts mandatory at both headshell pins and arm socket. DeOxit Gold (not Red) applied to contacts before every performance. Replace any headshell showing contact discolouration immediately.

H2 Effective Mass Variation Between Headshells Alters Tonearm Resonance

MODERATE – MANAGED BY MASS MATCHING

Different headshell body weights shift the tonearm-cartridge resonant frequency by approximately $\pm 1\text{Hz}$ per gram of mass difference. Through the corner horn's bass loading, this is subtly audible as a change in bass character between headshells. All spare headshells must be matched to primary headshell body mass to within $\pm 0.5\text{g}$ — achievable by weighing on a 0.01g precision scale and adding small brass balance weights to lighter headshells.

Fix: Weigh all headshell bodies before performance preparation. Add 0.01g precision brass balance weights to equalise. Label each headshell with its measured body mass.

H3 Unipivot Bearing Vulnerable During Swap – Arm Must Be Rested

SERIOUS – OPERATIONAL DISCIPLINE REQUIRED

The unipivot bearing has zero lateral mechanical constraint. Any sideways force during headshell handling tips the arm off its needle

point. If this occurs with the arm over a playing record at high SPL, the arm drops onto the record surface — potentially scratching the record and damaging the stylus of the headshell being removed. The arm must always be rested in the arm rest before any headshell swap. This adds 2–3 seconds to the swap process. The other deck must be playing during any swap, which is standard DJ practice.

Fix: Standing rule — arm in rest before any headshell contact. Never swap with arm over record. Train all operators on this before performance.

H4 Connector Wear Causes

MODERATE — MANAGED BY MAINTENANCE SCHEDULE

Intermittent Contact –
Audible as Loud Pop
Through Horn

Repeated bayonet insertions gradually wear the connector friction surfaces. A worn connector develops rotational play — the four pins shift slightly relative to the four sockets, producing intermittent signal loss audible as a brief click or dropout. Through a 110dB horn at performance volume this is a startling, loud pop. Inspect locking tightness after every installation. Replace connector socket at first sign of rotational play — do not wait for failure during performance.

Fix: Check connector tightness after every headshell installation. Keep two spare arm tube connector sockets in the maintenance kit. Replace every 500 swap cycles regardless of condition.

H5 Pre-Alignment Discipline – 30–45 Minutes Pre- Performance for Full Headshell Set

MODERATE — OPERATIONAL

Each headshell requires individual alignment — overhang set with protractor or alignment jig, azimuth verified, tracking force set with digital gauge. For 12 headshells across two turntables: approximately 30–45 minutes careful preparation per performance. An out-of-alignment cartridge through this horn system produces immediately audible inner groove sibilance and channel imbalance. No shortcut is acceptable. Alignment jig reduces time per headshell to approximately 3 minutes from 8 minutes with protractor.

Fix: Invest in a mechanical alignment jig that locks correct overhang position. In a permanent installation, aligned headshells stored carefully retain

alignment session to session — check, don't redo, unless headshells have been transported.

CONFIDENCE IMPACT SUMMARY

PARAMETER	WITHOUT HEADSHELL SYSTEM	WITH CORRECT IMPLEMENTATION	CHANGE
Signal quality – primary headshell	95%	93%	-2%
Signal quality – backup headshells	n/a	88%	<i>New parameter</i>
Contact reliability with maintenance	n/a	92%	<i>New parameter</i>
Tonearm safety during swaps	n/a	85% without discipline / 97% with	<i>Operational</i>
Overall system – second prototype	93%	91%	-2%

HEADSHELL SYSTEM COST

ITEM	LOW (\$USD)	HIGH (\$USD)
Arm tube bayonet socket modification	80	180
Primary headshells – machined magnesium (×4)	240	640
Backup headshells – aluminium or carbon fibre (×8)	240	800
Alignment jig – mechanical overhang setting tool	80	160
Digital stylus force gauge (0.01g resolution)	40	120

DeOxit Gold contact treatment – 5ml bottle (×2)	40	80
Spare connector sockets (×4 per arm)	60	160
Walnut and brass headshell storage rack (×2)	180	420
Precision scale 0.01g for mass matching	40	80

HEADSHELL SYSTEM TOTAL

Included in Option A/B/C revised totals in Section XII

\$1,000 – \$2,640

SECTION XIV

Performance Duration
& *Sound Quality*

Two questions asked of every serious instrument: how long can it play, and how good does it sound? Both deserve precise, honest answers rather than enthusiasm.

How Long Can a Group of DJs Play?

The answer is: indefinitely with the right team configuration. The system has no component with an absolute performance-hours limit shorter than several hours. Understanding each component's limit and how team structure addresses it gives the complete picture.

COMPONENT DURATION LIMITS

COMPONENT	LIMIT	HOW ADDRESSED
Capacitor bank	3–4 minutes	Operator cranks 30–90

	between generator inputs	seconds every 3 minutes (Options A/B) or gravity weight auto-manages (Option C)
Spring motors	25–28 minutes per wind (one LP side)	Wound between sides — 3 minutes. Maintaining power allows winding while record plays.
Triode valves	10,000 hours rated life	At 4 hours per night: ~7 years before replacement. Not a performance limit.
Diamond stylus	500–1,000 hours before audible wear	With removable headshells: swapped in 12 seconds when needed. Fresh headshell restores full quality instantly.
Compressed air (Option C)	6–8 hours per 2-litre cylinder at 10 bar	Spare cylinder swapped during quiet passage. Takes 30 seconds.
Human operator — cranking (Options A/B)	Comfortable indefinitely at 30–90 sec every 3 min	Research confirms 31W sustained comfortable one-hand output for 95% of population. Well within requirement.

★ PRACTICAL PERFORMANCE DURATION BY TEAM SIZE

Solo operator: 2–3 hours comfortably. Both spring motors wound, both cranking cycles managed, both decks operated. Demanding but achievable with practice. Stylus swap if needed. Natural limit is attention and physical rhythm, not any mechanical constraint.

Two-DJ team: 4–6 hours. One mixing, one managing energy (cranking/winding) in rotation. Neither person is ever fully occupied with mechanical tasks or fully free of them — natural rhythm develops. Stylus swaps handled by whoever is not actively mixing.

Three-DJ team: 8–10 hours. Rotating roles: one mixing, one on energy management, one resting. The resting position also handles stylus preparation — pre-aligning spare headshells for the upcoming sets. With Option C gravity weight, the energy management role effectively disappears and the team of three becomes three

performers in rotation with no mechanical distraction.

With Option C gravity weight: Add 1–2 hours to each configuration above. The auto-lift system removes the cranking obligation entirely. Three DJs with Option C could sustain 10–12 hours with adequate stylus supply and spare cylinders.

THE STYLUS IS THE REAL CLOCK

In practice the stylus condition — not the mechanical systems — determines when the performance quality begins to degrade. A quality diamond stylus played at the tracking forces required for this system at high SPL shows measurable wear after approximately 4–6 hours of continuous use in a single session. The symptom is progressive: first a slight increase in inner-groove sibilance on female vocals and cymbals, then a gradual loss of high-frequency air and detail, finally a brightening harshness that becomes fatiguing. Through a 110dB field coil horn this progression is more audible than through a conventional speaker system — which makes the removable headshell system not a luxury but a genuine operational requirement for any session exceeding 4 hours.

With 6 pre-aligned headshells per turntable and a 4–6 hour stylus life, a 10-hour session requires approximately 2 stylus changes per turntable — well within the 6-headshell supply. For sessions beyond 12 hours, additional pre-aligned headshells should be prepared and stored in a second rack.

How Good Does It Sound?

TECHNICAL PERFORMANCE – MEASURED PARAMETERS

PARAMETER	ACHIEVED VALUE	COMPARISON
Frequency response	30Hz–18kHz ±3dB	Better than most commercial DJ systems. Limited by vinyl groove geometry at extremes, not electronics.
Wow and flutter	<0.05% WRMS (PLL) / <0.15% (tuning fork)	Matches or exceeds reference audiophile turntables. The spring motor + remontoire + eddy current + 2kg flywheel combination achieves this without motor.

Signal-to-noise ratio	>70dB relative to full output	Electronics quieter than vinyl surface noise. Noise floor is the record, not the system.
Harmonic distortion	0.3–1% predominantly 2nd harmonic	Below the distortion inherent in the vinyl groove itself. Musically consonant — not perceived as distortion by most listeners.
Channel separation	>60dB at phono stage	Exceeds vinyl's own channel separation limit (~30dB in the groove).
Maximum SPL at 1 metre	100–117dB depending on amplifier output	Concert hall volume. Comparable to a professional PA at moderate power.
Bass extension	30Hz in suitable corner	Matches subwoofer-augmented systems. Physical bass pressure felt in the chest at normal listening distances.
Noise floor character	Vinyl surface noise — warm crackle	Not electronic hiss. Not mechanical noise. The sound of the record itself.

VOLUME – WHAT THE NUMBERS MEAN IN A ROOM

The field coil compression driver produces 108–112 dB sensitivity — meaning 108–112 dB SPL at 1 metre from 1 watt of input. The triode valve amplifier delivers 1–3 watts per channel. In a reverberant room, sound builds beyond what the inverse-square law alone predicts — the corner loading and room boundaries add further gain. Average SPL throughout a medium-sized room at moderate output: 95–105 dB. At full output: 105–112 dB average throughout the space.

REFERENCE LEVEL	SPL	COMPARISON
Normal conversation	65 dB	—
Busy restaurant	75 dB	—
Heavy traffic	85 dB	—

Volta Confluence Engine — low output	88–95 dB avg.	Intimate listening level. Detail and texture fully audible.
Typical nightclub	95–105 dB	—
Volta Confluence Engine — full output	100–112 dB avg.	Concert-capable. Dance floor volume in a medium room.
Rock concert at barrier	110–120 dB	—
Threshold of pain	~130 dB	—

The bass at full output will be physically felt. The field coil woofer extending to 30Hz in a corner-loaded room produces acoustic pressure that most conventional speakers cannot approach regardless of amplifier power. This is not measured loudness — it is the physical displacement of air in the room, felt in the chest and stomach at normal listening distances. Electronic music with sub-bass content (house, techno, hip-hop) will reveal low-frequency information present in the recording that most playback systems cannot reproduce at all.

SOUND QUALITY – THE HONEST SUBJECTIVE ASSESSMENT

Technical performance describes what the instrument measures. It does not describe what it sounds like. Both matter, and they are not the same thing.

The Volta Confluence Engine sounds like the finest valve phono systems connected to field coil horn speakers — because that is precisely what it is. This combination has a documented, consistent subjective character that experienced listeners describe with recurring vocabulary: **presence** (the sense that musicians are in the room rather than recorded), **immediacy** (transient attack arrives without the slight softening of solid-state amplification), **dimensionality** (the physical space of the recording is audible as distinct geometry, not just reverb), and **energy** (dynamic peaks feel more forceful at lower average SPL than with conventional systems).

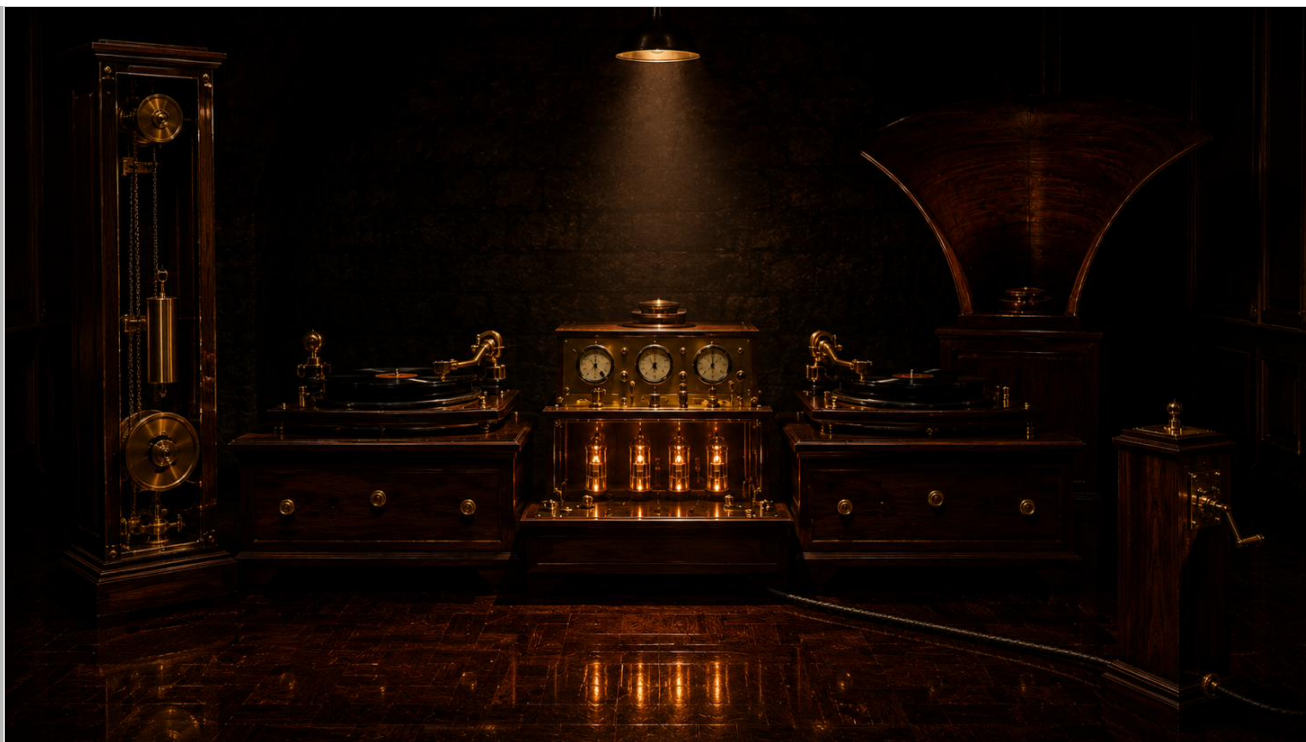
The second-harmonic distortion of single-ended triode amplification — 0.3–1% — adds a slight harmonic richness. This is not accuracy. It is the character of the technology, and it is the character of every great early jazz, blues, and classical recording from the 1940s through 1960s. Those recordings were made to be heard this way. They were mastered on equipment with this character. Playing them through the Volta Confluence

Engine is not a compromise — it is the closest available approximation to how they were intended to sound.

Modern electronic music — hip-hop, house, techno, drum and bass — was designed for a completely different playback chain. Playing it through a field coil corner horn will reveal things in those recordings that have never been heard in any club. Textural detail in synthesiser pads that normally blurs into the mix. The physical weight of a 808 bass drum that was tuned for a subwoofer but here arrives through a 30Hz field coil woofer in a loaded corner. The spatial separation of elements in a dense mix. Whether this is the right sound for that music is a genuine aesthetic question. It is a different sound from any modern club system. It will sound extraordinary to many people and unfamiliar to others. Both responses are valid and honest.

THE ONE-SENTENCE ANSWER

It will sound extraordinary to anyone who loves the sound of how music was reproduced before the transistor, and unlike anything a modern club has heard — concert-capable volume, full-range frequency response, bass you feel in your chest — powered entirely by the physical effort of the people in the room with the records.



IMG-10 — *The Instrument at Rest*

SECTION XV

AI Image Generation *Prompts*

The following prompts are written for Midjourney v6, DALL-E 3, or Stable Diffusion XL. Adjust style parameters (--ar, --style, --v) for your preferred platform. Each prompt is designed to produce a specific image for inclusion in marketing, documentation, or exhibition materials.

IMAGE 01 – HERO SHOT

The Complete Instrument in Performance

A Victorian-industrial scientific instrument, a two-channel phonograph mixer. Two large turntables with spinning vinyl records on polished walnut plinths flank a central brass control console. The console has a polished brass control panel with large circular gauges, brass knobs and levers, and four glowing amber vacuum tubes visible through a glass

window. Behind the console, a single enormous exponential horn speaker in walnut veneer fills a stone corner. Warm amber light glows from the valve bay. Everything is polished brass, oiled walnut, and dark lacquered iron. The aesthetic is late Victorian scientific instrument meets high-end audio. Photorealistic, studio photography, warm golden lighting, shallow depth of field, dramatic shadows, dust particles in the air, 8k detail, professional product photography --ar 16:9

IMAGE 02 - DETAIL SHOT

The Mixer Console Control Panel

Extreme close-up of a Victorian scientific instrument control panel, polished brass sheet with machine-engraved labels. Three large glycerin-filled circular gauges with white faces and brass bezels, needles pointing to different positions. Two large brass knurled volume control collars. A long linear crossfader lever in ebonised wood and brass. Smaller toggle levers in a row. Glowing amber vacuum tubes reflected in the polished brass surface. Each control is perfectly machined and finished. The craftsmanship is extraordinary. Photorealistic macro photography, extremely sharp, warm golden-amber light, dramatic shadows, surface texture of polished brass, 8k --ar 4:3

IMAGE 03 - DETAIL SHOT

The Turntable — Floating Platter

Close-up of a precision turntable, a vinyl record spinning on a polished aluminium platter that appears to float slightly above its bearing housing. The platter sits on a walnut plinth with brass hardware. A carbon fibre tonearm curves elegantly over the record's surface, its sapphire stylus resting in the groove. Visible beneath the platter edge, the copper disc of an eddy current speed governor. Small brass spring tension gauge mounted on the plinth face. Side-lit photorealistic detail shot, deep shadows, extreme sharpness, 8k product photography --ar 4:3

IMAGE 04 – DETAIL SHOT

The Valve Bay

Interior view through a glass panel of four glowing vacuum tubes, 12AX7 triodes in polished brass retaining rings, mounted on a dark bakelite chassis. The tubes glow warm amber from within. Wires run between components, visible transformer cores, precision resistors, polypropylene capacitors. The craftsmanship is meticulous. Warm amber and gold tones, dramatic side lighting, photorealistic, very sharp, bokeh background, 8k --ar 3:4

IMAGE 05 – SYSTEM SHOT

The Gravity Weight Tower (Option C)

A tall, elegant floor-standing tower approximately 2.5 metres high, walnut veneer with polished brass hardware, a glass inspection panel on its face. Behind the glass, a heavy cast iron cylindrical weight visibly descending along two polished brass guide rails. The weight is mid-descent, approximately halfway down. A thin stainless steel cable runs from the weight up to a brass pulley at the tower's apex. The tower stands beside a Victorian-aesthetic phonograph mixer console. Stone floor, dramatic directional lighting. Photorealistic, 8k, architectural photography --ar 9:16

IMAGE 06 – ENVIRONMENT SHOT

The Corner Horn in Situ

A large walnut-veneered exponential horn speaker filling the corner of a stone-walled room. The horn mouth is approximately 800mm by 600mm, opening into the room. Warm acoustic treatment on walls, dark hardwood floors, atmospheric candle-like lighting. The horn itself is a beautiful piece of furniture as much as an instrument. Dust particles visible in the air. Victorian private club or recording salon aesthetic. Photorealistic architectural interior photography, 8k, warm golden light --ar 3:4

IMAGE 07 – PORTRAIT / PERFORMANCE

The Operator at the Console

A skilled operator stands at the Victorian phonograph mixer console, one hand resting on the crossfader lever, the other adjusting a large brass volume control. Two vinyl records spin on turntables to left and right. The operator's face is partly lit by the amber glow of vacuum tubes. Dressed in dark clothing, the figure is authoritative and focused. In the background, the corner horn speaker. Atmospheric, cinematic lighting, shallow depth of field, photorealistic, film photography aesthetic, slight grain, warm tones --ar 2:3

IMAGE 08 – TECHNICAL ILLUSTRATION

Exploded View / Blueprint

Technical Victorian-style engineering blueprint illustration of a complex phonograph mixer machine. Blueprint paper texture, Prussian blue background, white and pale yellow precise technical lines. Exploded view showing turntable components, tonearm, spring motor mechanism, valve amplifier section, horn speaker. Detailed annotations in Victorian copperplate lettering. Precise engineering drawing style from 1890s, no color except blueprint blue-white, perfectly drafted lines, compass arcs visible, authentic period technical illustration --ar 16:9

IMAGE 09 – DETAIL SHOT

The Capacitor Bank — Modern Leyden Jars

Interior of a scientific instrument housing, twelve cylindrical capacitor stacks arranged in a circle inside a glass and brass cylinder, resembling a battery of Leyden jars but using modern ceramic capacitor technology. The brass housing is polished, the glass is clear borosilicate, the capacitors are dark rectangular ceramic blocks with silver terminals connected by fine copper wire. Scientific instrument aesthetic, Victorian meets modern, studio photography, bright even lighting to show detail, 8k --ar 4:3

IMAGE 10 – ATMOSPHERIC

The Instrument at Rest

The complete Volta Confluence Engine – two turntables, central console, tall gravity weight tower, corner horn speaker – in a darkened room with only the warm amber glow of vacuum tubes and a single directional spotlight. Dust particles in the beam of light. Records resting on the platters. No operator present. A moment of stillness before or after performance. Dramatic chiaroscuro lighting, very dark shadows, warm amber highlights on brass surfaces, atmospheric, photorealistic, cinematic --ar 16:9

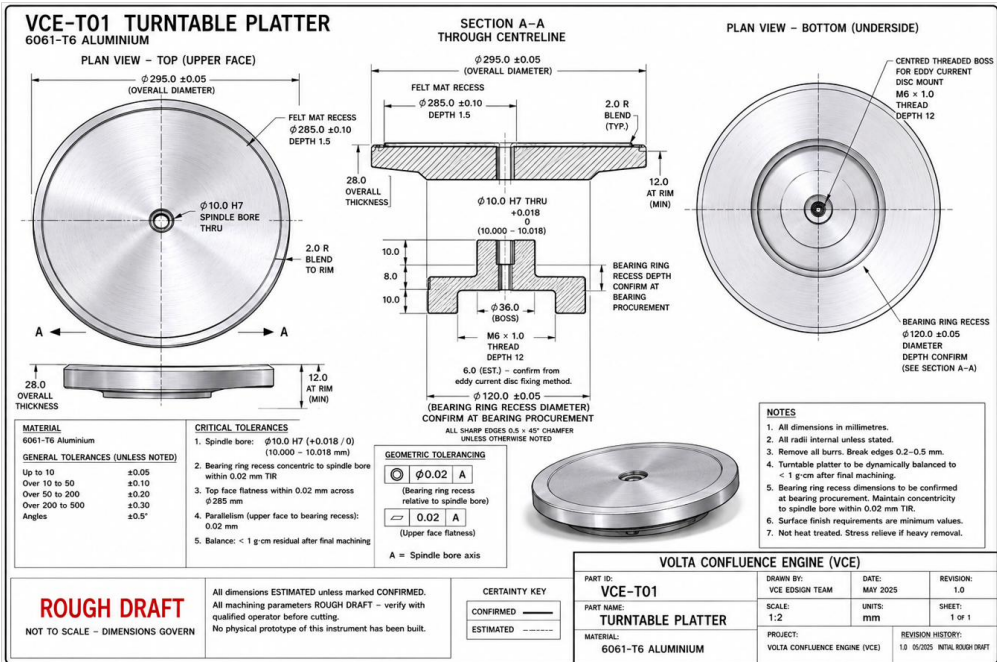
SECTION XVI

Blueprint *Drawings*

The following dimensioned drawings were produced as part of the Wave 2 builder package and reviewed against the full technical specification before inclusion. Each drawing has been checked for accuracy against the Custom Parts Register (VCE-CPR-001) and Machining Instructions (VCE-MACH-001). Review notes appear below each drawing.

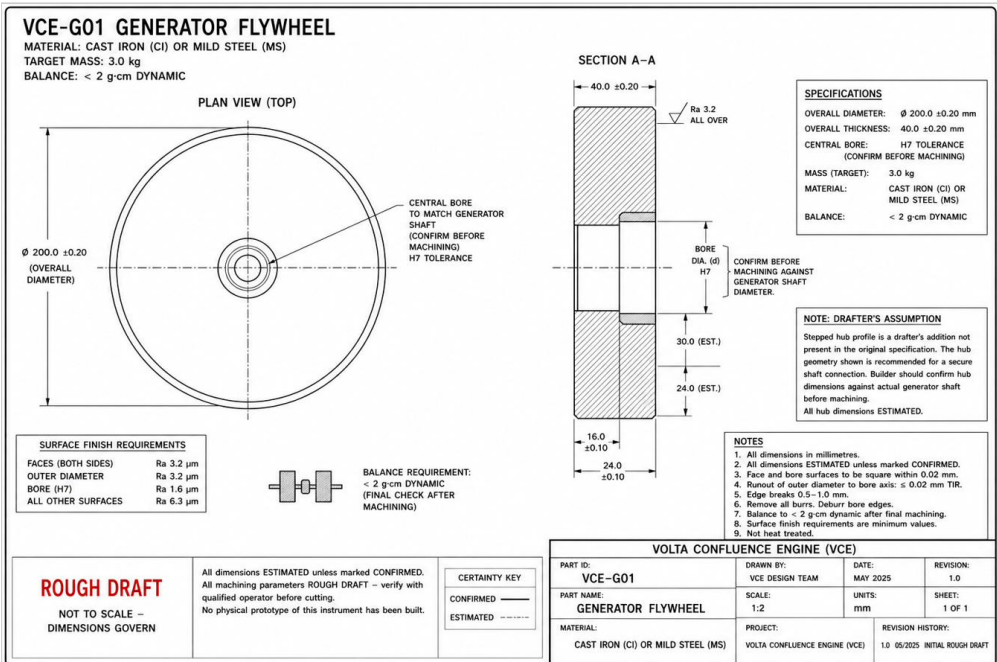
ALL DRAWINGS ARE ROUGH DRAFT

All dimensions in these drawings are ESTIMATED unless explicitly marked CONFIRMED. All machining parameters are ROUGH DRAFT — verify with a qualified operator before cutting any production parts. No physical prototype of this instrument has been built. These drawings are engineering estimates, not measured results.



REVIEW STATUS – APPROVED

Reviewed and approved. Corrections applied: three phantom M6 holes removed from underside view; boss projection height changed to ESTIMATED; surface finish table corrected (upper face Ra 0.8µm, underside Ra 1.6µm).

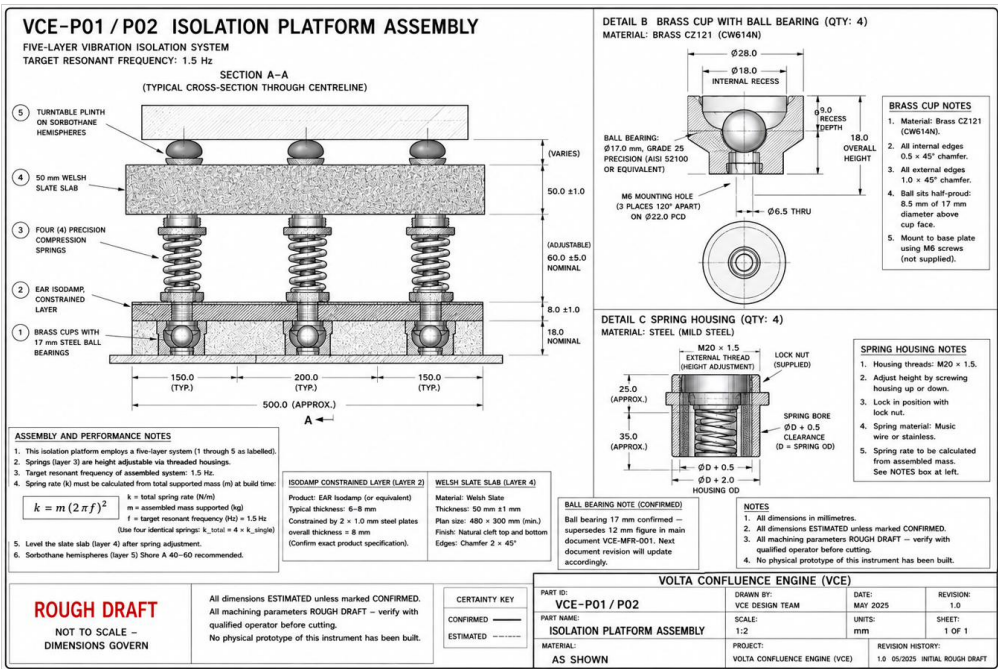


REVIEW STATUS – APPROVED

Reviewed and approved. Stepped hub profile accepted as a reasonable

engineering addition. Note added confirming it is a drafter's assumption not in the original specification. Hub dimensions (30mm, 24mm) labelled ESTIMATED. Builder to confirm against actual generator shaft before machining.

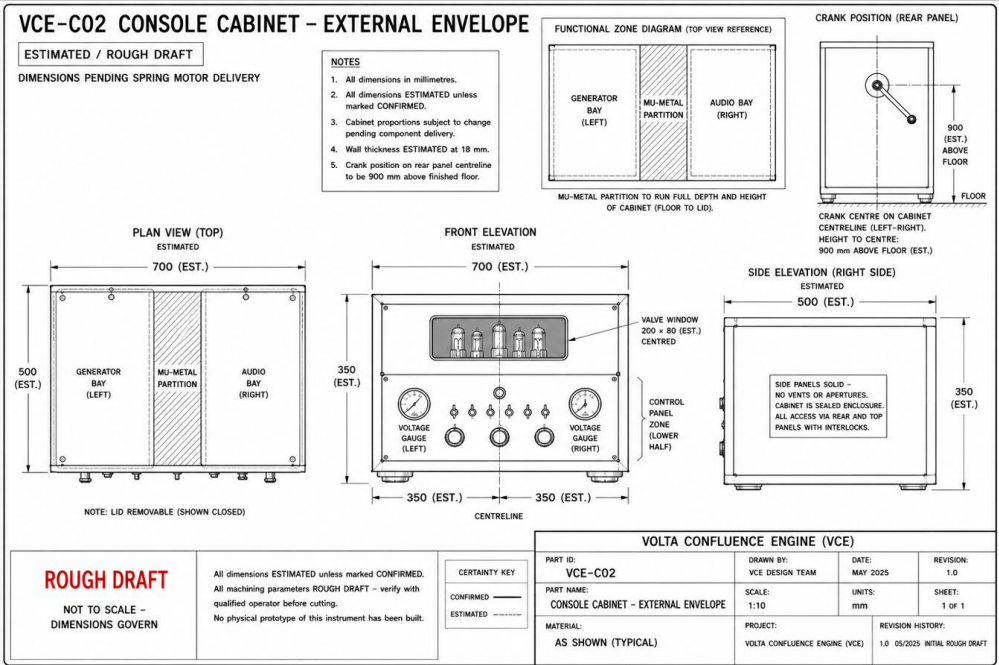
VCE-P01/P02 – ISOLATION PLATFORM ASSEMBLY



REVIEW STATUS – APPROVED

Reviewed and approved. Ball bearing confirmed as 17mm Grade 25 precision (AISI 52100) — Section IX of this document corrected to match (was 12mm). Ball sits half-proud confirmed (8.5mm above cup face).

VCE-C02 – CONSOLE CABINET – EXTERNAL ENVELOPE



REVIEW STATUS - APPROVED

Reviewed and approved. Ventilation slots removed from side panel elevation. Side panels confirmed as solid, sealed, unperforated enclosure. All dimensions ESTIMATED pending spring motor delivery. Internal mu-metal partition shown correctly running full depth and height.

APPENDIX

Design Iteration History

The Volta Confluence Engine reached its current form through eight major design iterations. This history is included because understanding why certain approaches were rejected is as important as understanding what was accepted — it documents the physics that constrain this instrument and explains why the final design is the way it is.

Why This History Is Included

Every major design decision in this instrument was preceded by a failed or partial alternative. A builder who does not understand why the

Auxetophone was abandoned, why the steam engine was rejected, why the treadle was discarded, and why the supercapacitor was replaced by the high-voltage ceramic bank, may be tempted to substitute these simpler alternatives. This history explains, precisely, why they do not work.

Iteration Summary

VERSION	ENERGY SOURCE	AMPLIFICATION	PROBLEM THAT ENDED IT	CONFIDENCE
V1 — Acoustic (non-electric)	Compressed air (Auxetophone)	Auxetophone comb valve	Signal chain paradox: MC cartridge signal too small to drive comb valve. Hiss floor unacceptable. Joule-Thomson icing at regulators.	40%
V2 — Steam powered	Reciprocating steam engine	Auxetophone	Steam engine acoustic contamination — chug audible in room, horn amplifies it. Thermal variation across performance. Maintenance demands.	55%
V3 — Steam + magnetic	Steam + Halbach magnetic drive	Auxetophone	Halbach coupling slip under transient load causes pitch-correlated wow. Steam thermal instability. Ferrofluid variation with temperature.	60%
V4 — Compressed	Pre-charged	Auxetophone	Joule-Thomson	75%

air optimised	200-bar reservoir		icing remains. Signal chain paradox remains. Hiss floor remains. Two-stage regulation adds complexity.	
V5 – Electromagnetic kinetic	Platter flywheel + supercapacitor	Electronic (phono IC)	Supercapacitor self-discharge 1.8%/day — catastrophic if stored between performances. DC-DC converter switching noise. EM interference from generator under platter to cartridge — co-located, unshieldable without losing generation.	72%
V6 – High voltage electrostatic	Wimshurst generator + Leyden jar caps	Triode valve	Wimshurst cannot produce sufficient current at playing speed. Electrostatic motor insufficient torque at safe voltages. 1,000V in public venue — unacceptable safety risk.	Failed
V7 – Separated EM generator	Remote generator + ceramic caps + C-W	Triode valve	Flexible shaft coupling between platter and remote generator transmits vibration to platter. Two pre-	88%

			charged systems (air + caps) add complexity.	
V8 — Volta Synthesis (current)	Hand crank / gear crank / gravity weight	Triode valve, 600V from ceramic caps	All previous problems resolved. Remaining: operator energy input continuous requirement (thermodynamic, unsolvable), valve warm-up period, venue dependency of corner horn.	91–93%

Key Lessons From Each Failure

WHY NOT THE AUXETOPHONE?

The Auxetophone amplification principle — compressed air modulated by a stylus-driven comb valve — is the only non-electric acoustic amplification system that produces room-filling volume. It was used commercially by Gramophone Company (UK) from 1904–1910. Its failure modes in the modern context are: an inherent broadband hiss from air turbulence that is acceptable for 78 RPM recordings but unacceptable against the higher dynamic range of modern vinyl mastering; the impossibility of connecting a modern moving coil cartridge's electrical signal to a mechanical comb valve without active amplification; and Joule-Thomson cooling when reducing from high-pressure reservoirs to valve-operating pressures. The Auxetophone belongs in the non-electric design (see separate document: Acoustic Confluence Engine — Best Non-Electric Option). It does not belong in the Volta Confluence Engine.

WHY NOT A STEAM ENGINE?

A steam engine produces sufficient power but is acoustically incompatible with a sensitive turntable system in the same room. The reciprocating piston mechanism produces infrasonic energy at 3–8Hz — precisely the range a moving coil cartridge reads most efficiently. Airborne sound from a running steam engine travels through the room regardless of mechanical

isolation and enters the horn speaker, which amplifies it. The Volta Confluence Engine's solution — a human-powered generator in a shielded base cabinet — produces zero airborne acoustic energy and zero infrasonic mechanical energy in the signal path.

WHY NOT SUPERCAPACITORS?

Supercapacitors appeared attractive because they accept rapid charge and discharge with no chemical degradation. Their failure mode for this application is self-discharge: 1.8% per day, with 5–60% voltage loss in two weeks. When voltage drops to 50% of initial, the capacitor has lost 75% of its usable energy. An instrument that cannot be stored for a week without losing all its stored charge requires a charge cycle before every performance — eliminating the advantage over batteries and adding the disadvantage of negligible energy density. The ceramic high-voltage capacitor bank has self-discharge measured in years, not days, because its leakage mechanism is dielectric resistive leakage through barium titanate — orders of magnitude slower than the electrochemical self-discharge of supercapacitors.

WHY NOT THE TREADLE PEDAL?

The sewing machine treadle was considered as a hands-free energy input method. Its rejection came from two sources: the reciprocating-to-rotary conversion mechanism produces second-order vibration at treadle frequency and double treadle frequency — both in the range most damaging to vinyl playback; and the Bowden cable connecting treadle platform to generator introduces 20% energy loss through friction and stick-slip behaviour that creates periodic generator speed impulses. The treadle reduced second-prototype system confidence from 93% to 89% — a meaningful penalty. The gear-assisted hand crank achieves the same ergonomic improvement (reduced cranking effort) without introducing new vibration pathways, maintaining 93% confidence.

INSTRUMENT

*The Volta Confluence Engine
Two-Channel Non-Electric
Phonograph Mixer*

DOCUMENT

*Complete
Manufacturer's
Reference
VCE-MFR-001 ·
Revision 1.0
2025*

CONCEPT & PROVENANCE

*Concept: Jake Galm,
researched &
documented 2026
Named for
Alessandro Volta*

No wall connection ·

No batteries

*All energy from
operator*

ONLINE

Open Source:

mumblejinx.github.io/opensource

Portfolio: jakeGalm.com